

iBlueDrive Control Manager user's guide

DOC No. M033O03 - December 2019

Description of iBlueDrive free control software for iBlueDrive devices. Main features and how to use them for configuring parameters.

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1. What is the iBlueDrive Control Manager?

iBlueDrive Control Manager is the free software property of DCM Sistemas designed to control and set up all iBlueDrive technology devices. iBlueDrive Control Manager (henceforth iBlueDrive CM) does not need any previous setup to start working. Once the software is installed, you can run the application and use it.

iBlueDrive lighting systems are completely factory configured to work automatically but, if necessary, this software allows the user to configure all the parameters for the operation of the device to meet the needs of the application.

For connecting devices to PC and use this software, it is essential to use iBlueDrive Box (VTA0005A) or iBlueDrive USB (VTA0007A).

iBlueDrive Box (VTA0005A) allows to configure and monitor iBlueDrive lighting systems, 'online or offline', connecting at the same time up to 20 different lamps, working with the same signal trigger. Use it to change lighting settings or to control online the light configuration. The **iBlueDrive box** combines the trigger signal with the bidirectional data communication to allow changes while the light is working in strobe mode. You can also turn ON and OFF the light, change output power or any other settings by software.

iBlueDrive USB (VTA0007A) allows to configure iBlueDrive lighting systems, 'offline'. Use it to change lighting settings. The **iBlueDrive USB** allows to change output power, timing or any other settings by software.

The latest version of the software can be found on the DCM website (<https://www.dcmsistemas.com/en/products/accessories/software/>) where you can also find older versions if needed.

SOFTWARE SPECIFICATIONS & PREVIOUS CONSIDERATIONS

PC requirements	Windows Vista, W7, W8 or W10
Configuration backups	Yes
Updatable firmware	Yes

The usage process will be the following:

1. Download the package from our website (ZIP file). Last version will always be named like this: iBlueDriveCM.zip
2. Unzip the 'iBlueDriveCM.zip' file. You should access a folder with the following content. Read the instructions before starting.
3. In order to execute iBlueDrive CM:

Save files <devices.dat> and <iBlueDriveCM_1_4_0.exe> in the desired destination.

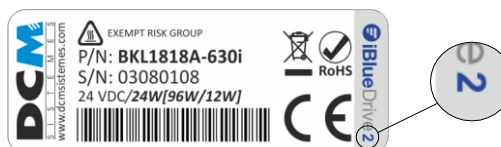
Windows must detect and install automatically the necessary drivers. iBlueDrive Box and iBlueDrive USB do not need any power supply for this step.

- If the driver has not been automatically installed from Windows update or you would like to install it manually, unzip the <CDM v2.12.28 WHQL Certified.zip> file and install it.
- Click on the <.exe> file and the program will start automatically. No installation is required.

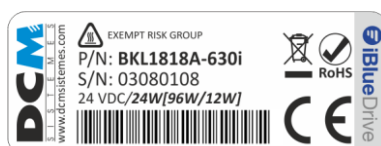
2. Overview

The iBlueDrive Control Manager software allows the user to manage and configure all the devices equipped with iBlueDrive technology:

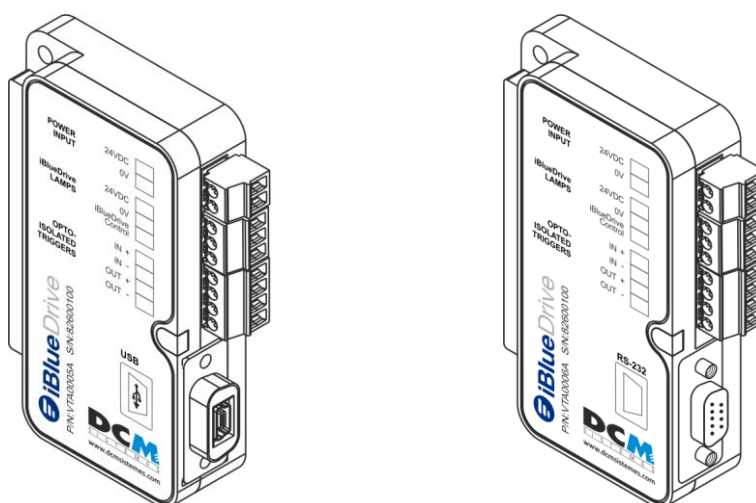
- iBlueDrive2 lightings. Are those devices with the following label:



- iBlueDrive lightings. Are those devices with the following label:



- iBlueBox devices (with USB or RS-232 connector):

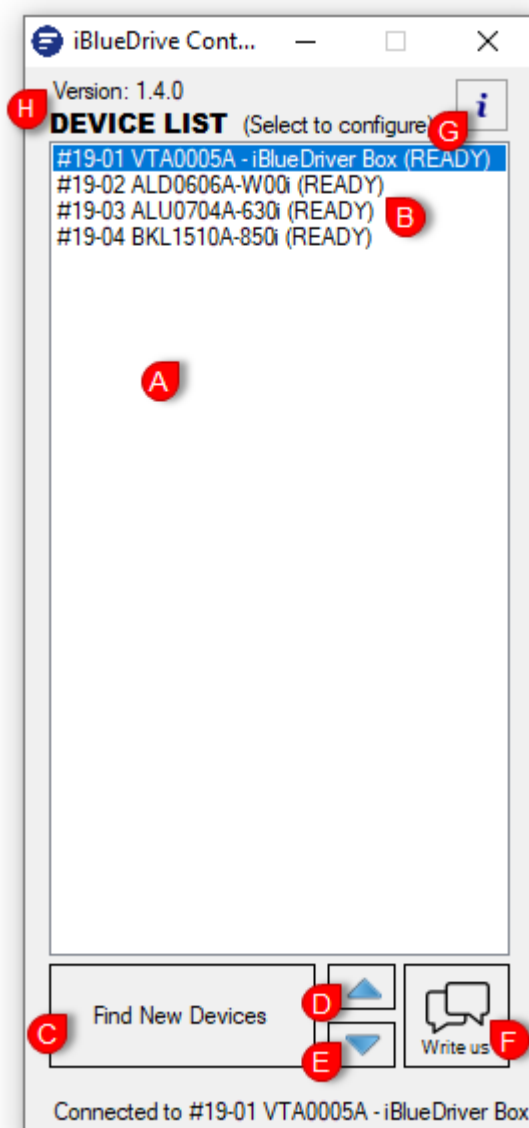


When the iBlueDrive Control Manager is started a main window that contains a list of the connected devices is shown. Once a device is detected its name appears on the list. When it is clicked, a side panel is opened, showing the features of that device. The panel will show the particular features of the device which has been clicked on, and are explained over the following sections of this document.

If you want to know more about iBlueDrive technology and electrical and technical specifications of the lightings please visit www.ibluedrive.com.

3. Main Window

This is the software appearance when it is executed and some devices are connected to the PC. In this case an iBlueBox and three lightings are connected and ready.



A. Device list

The device list shows the devices detected by the iBlueDriveCM. Detected devices will automatically be shown on it.

B. Device identifier

Each device in the list has a unique name according to the format **#Port-ID Device_name(Status)**

For example **#19-02 ALD0606A-W00i (READY)**, being:

- **Port:** is the COM Port to which the device is connected (19)
- **ID:** is the BUS identifier, needed for low-level communications (02)
- **Name:** is the product name (ALD0606A-W00i)
- **Status:** indicates the status of the device (READY). Other possible statuses are:
 - **LOADING:** the software is receiving data from the device.
 - **READY:** the device is ready to use.
 - **HW_UNKNOWN:** the device hardware version is not compatible with the current software version. You will be asked to update your software.
 - **FIRM_UNKNOWN:** the device firmware version is not compatible with the current software version. You will be asked to upgrade or downgrade the firmware.
 - **REF_UNKNOWN:** the product name is not recognised. You will be asked to update the file 'devices.dat'.
 - **UPDATING_FIRM:** the firmware of the device is currently being updated.

C. 'Find new devices' button

Clears all the devices in the Device List and starts a search through all the connected ports to detect all connected devices. If the software is not able to find any device, a message will be shown.

D. 'Up' button

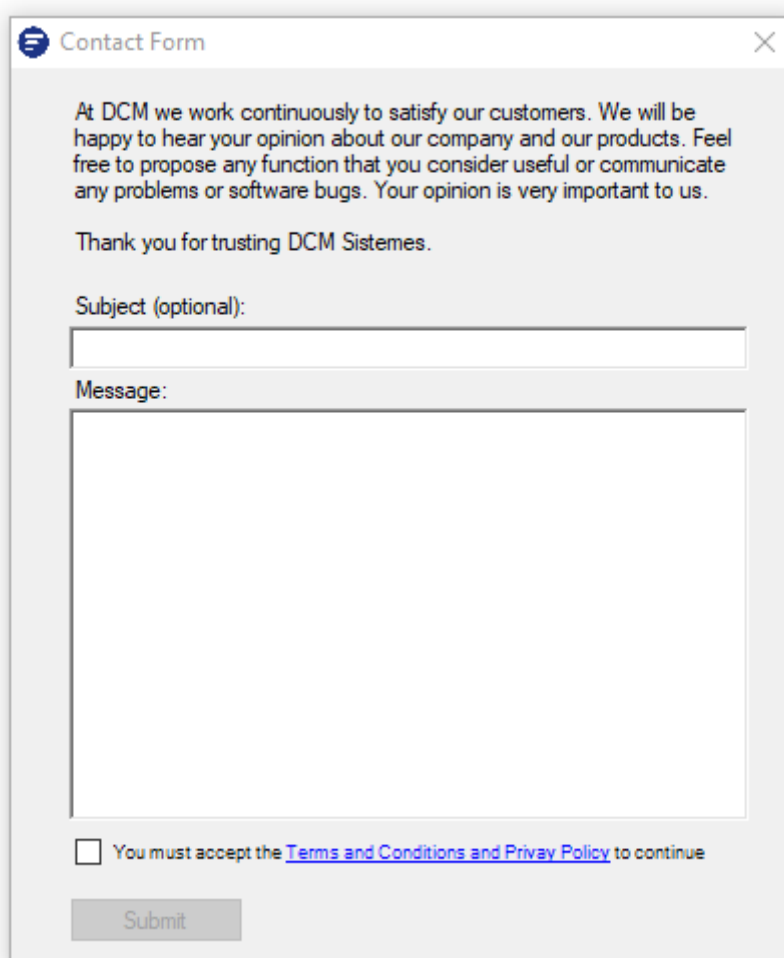
Increases the ID of the selected device, moving it down in the Device List

E. 'Down' button

Decreases the ID of the selected device, moving it up in the Device List

F. 'Write Us' button

Shows a form in which the user can submit any question or suggestion about the software operation to DCM Sistemas support team.



The image shows a 'Contact Form' dialog box with a title bar containing a menu icon and a close button. The form contains a welcome message, a 'Subject (optional)' text box, a 'Message' text area, a checkbox for terms and conditions, and a 'Submit' button.

Contact Form

At DCM we work continuously to satisfy our customers. We will be happy to hear your opinion about our company and our products. Feel free to propose any function that you consider useful or communicate any problems or software bugs. Your opinion is very important to us.

Thank you for trusting DCM Sistemas.

Subject (optional):

Message:

☐ You must accept the [Terms and Conditions and Privacy Policy](#) to continue

Submit

G. Info Button

Shows software version and disclaimer on a new window.



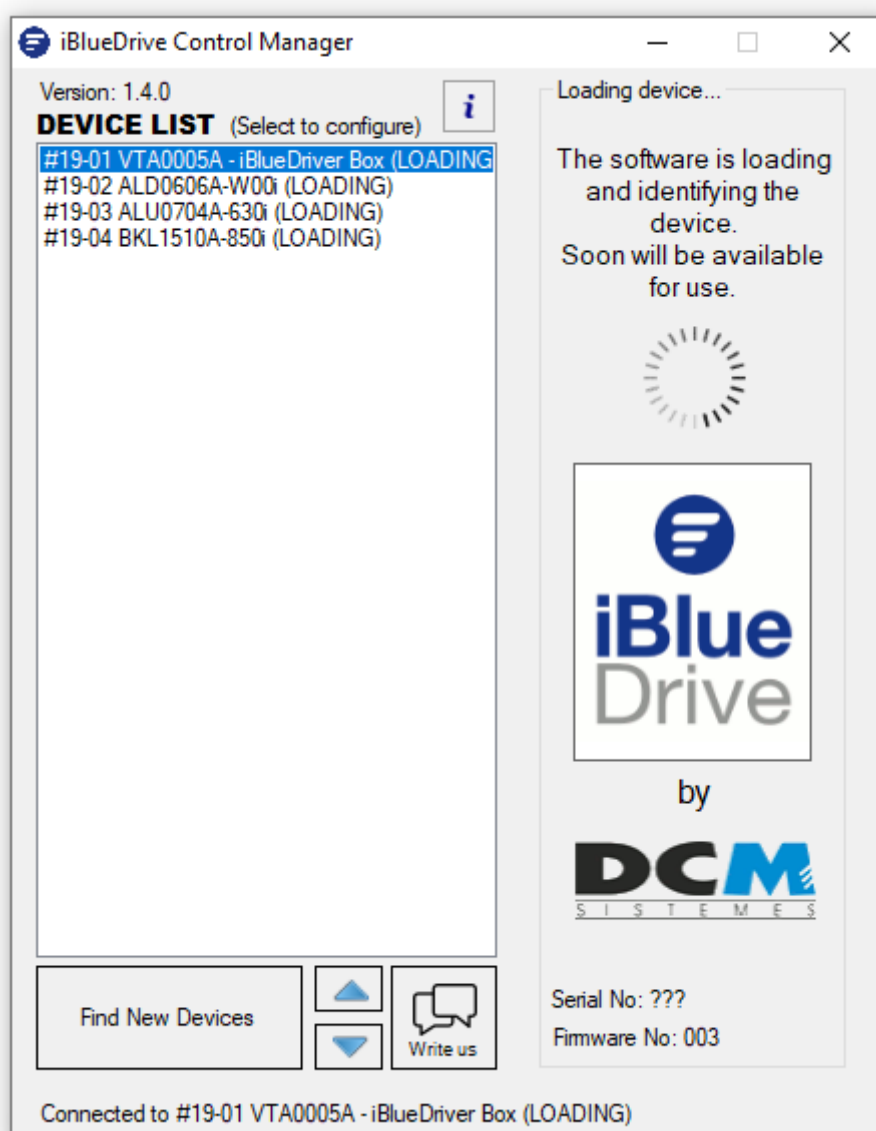
- Version news: Shows the new features added to the current software version.
- Close: Close the info window.

H. Software version

Shows the current software version.

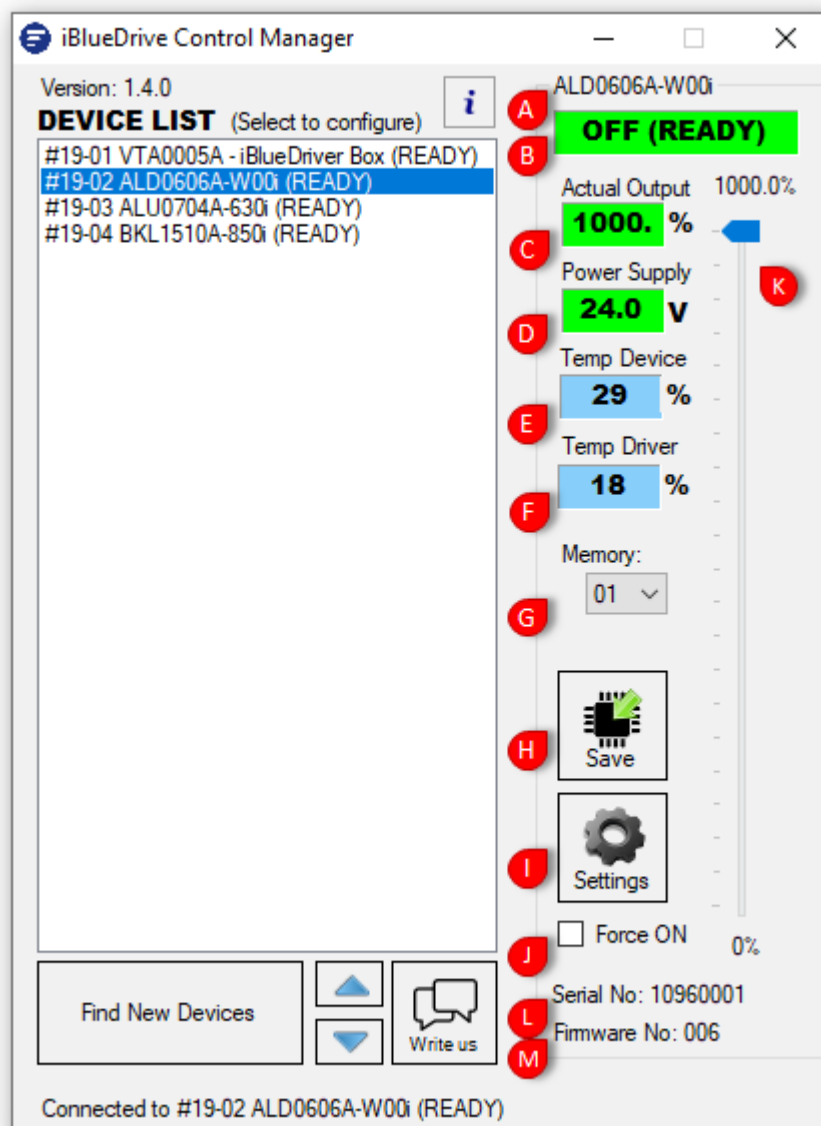
4. Loading View

This screen will be shown while the selected device is transmitting data to the software. It takes the software some seconds to read all the parameters from each connected device.



5. iBlueDrive2 lightings

5.1. Main view



A. *Product name label*

Shows the product name of the device

B. *Lighting Status*

- **OFF (READY):** the device light is off.
- **CONT.:** the device light is in continuous mode.
- **POWERED:** the device light is in powered mode.
- **FLASHING:** the device light is firing strobe pulses.
- **LOW VOLTAGE:** the input voltage is lower than 18.3V
- **OVERHEAT:** the driver temperature or the device temperature is higher than the operation temperature.

C. *Actual Output*

Lighting output power related to the maximum power in continuous mode, expressed as a percentage.

D. *Power Supply*

Voltage of the input power supply.

E. *Device temperature*

Temperature of the device related to its maximum operation temperature, expressed as a percentage.

F. *Driver temperature*

Temperature of the driver related to its maximum operation temperature, expressed as a percentage.

G. *Memory*

Indicates de memory bank currently in use. iBlueDrive lightings counts with 16 memory banks. Each memory bank can record one power output value.

H. *Save Button*

Saves the actual parameters configuration in the device's RAM memory. The button border turns red when some parameter's value has changed showing that the RAM memory is out of date. If the power supply is then disconnected before saving those changes will be lost.



I. *Settings Button*

Opens the side panel to operate the advanced settings

J. Force ON

When checked the lighting will turn on regardless the trigger input or the control signal.
When unchecked the lighting will turn on according to the trigger input or the control signal.

K. Power Slider

Controls the lighting output power.

L. Serial No

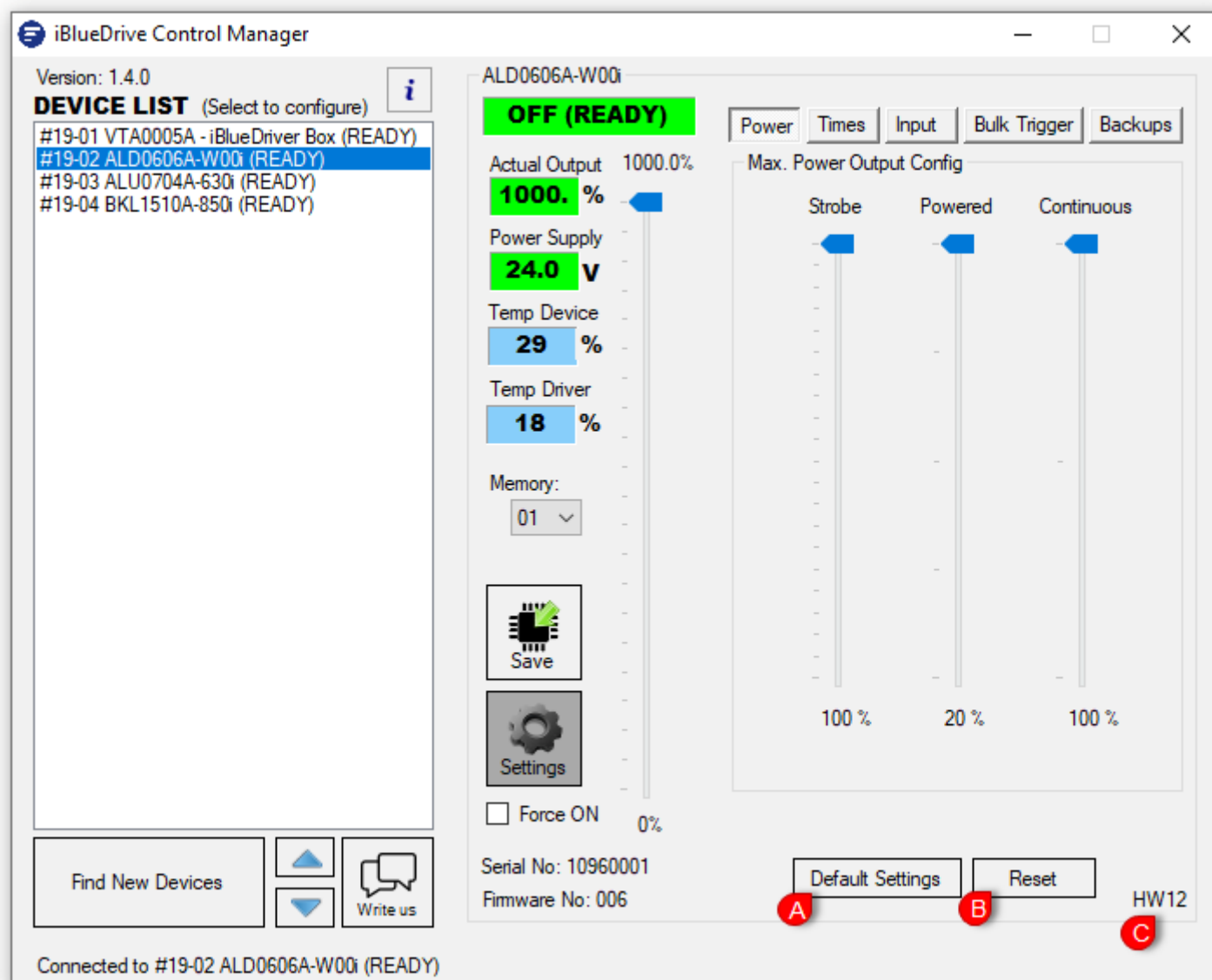
Shows the serial number. This number is unique for every single DCM device.

M. Firmware

Shows the firmware version of the device.

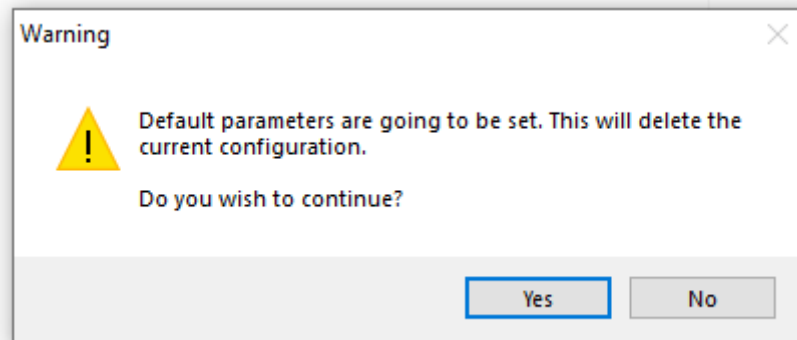
5.2. Settings

The configuration panel displays all configurable parameters of the device, allowing full adjustment to suit the needs of the application.



A. *Default Settings button*

Set all the parameters to their default value and save them in the device's RAM memory. A confirmation message will be shown before proceed.



B. *Reset Button*

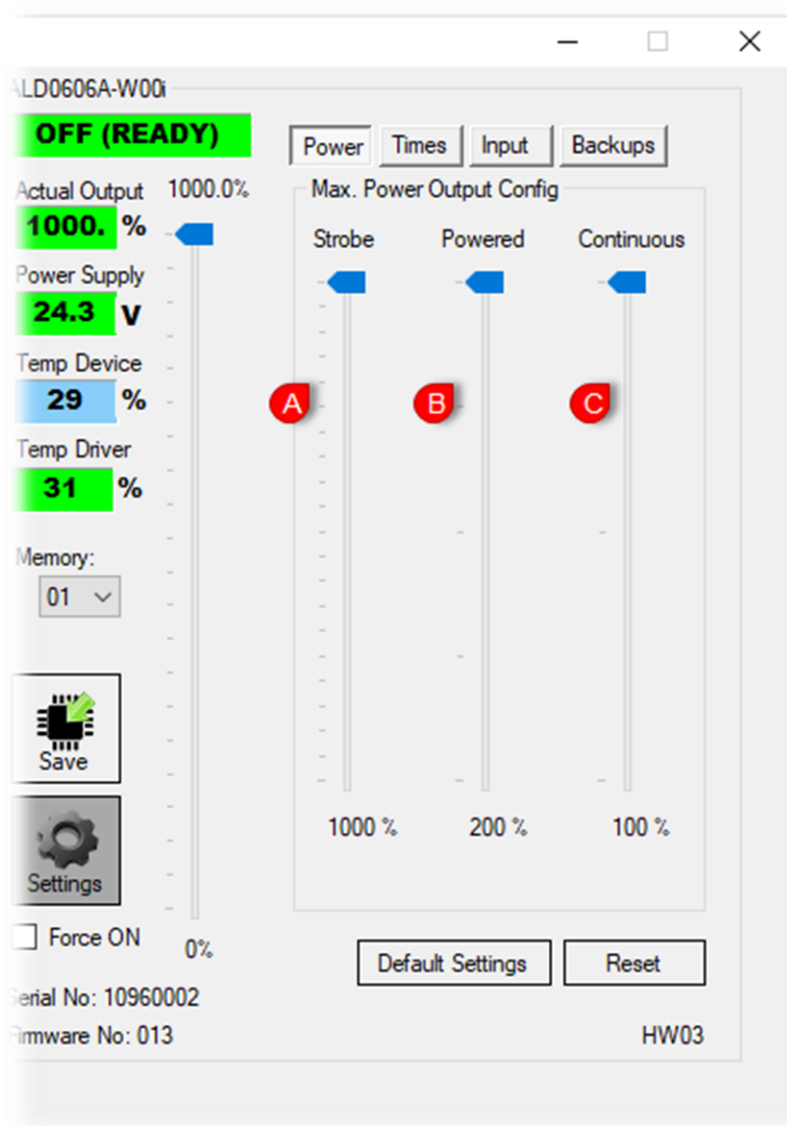
Performs a reset in the device. This equals to disconnect and then reconnect the device to the power supply.

C. *Hardware label*

Shows the hardware version of the device electronics. For the iBlueDrive2 is HW12.

5.2.1. Power tab

This tab shows the controls related to the output power of the device.



D. Strobe slider

Controls the device output power when working in strobe mode (FLASHING).

E. Powered slider

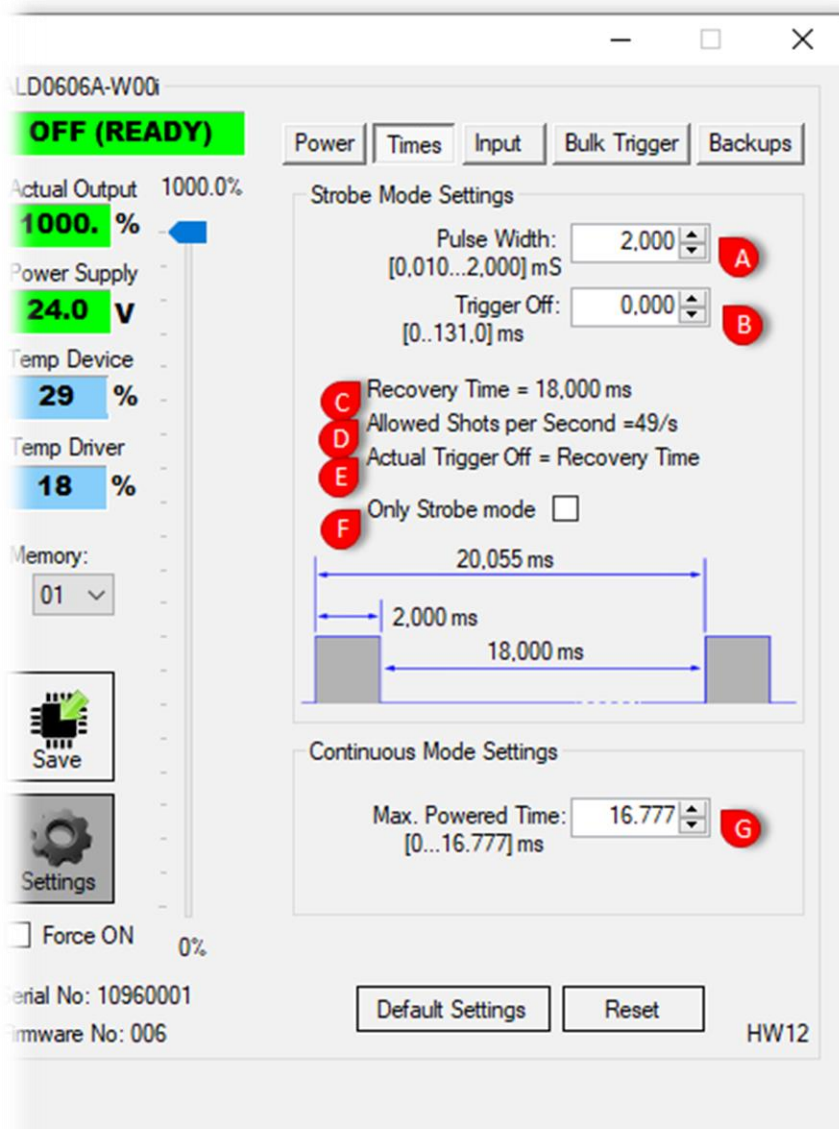
Controls the device output power when working in powered mode (POWERED).

F. Continuous slider

Controls the device output power when working in continuous mode (CONT.).

5.2.2. Times Tab

A diagram dynamically shows Pulse Width, Actual Trigger Off and the minimum time between strobe triggers.



A. Pulse Width

Allows you to change the light on time when operating in strobe mode (FLASHING).

B. Trigger Off

Allows you to customize the light off time after a strobe operation.

C. Recovery Time

Is the minimum time that the light must be off after a strobe operation. The recovery time will depend on the Pulse Width value.

The relation between Recovery Time and Pulse Width cannot be modified as it depends on technical and constructive parameters specific to each product.

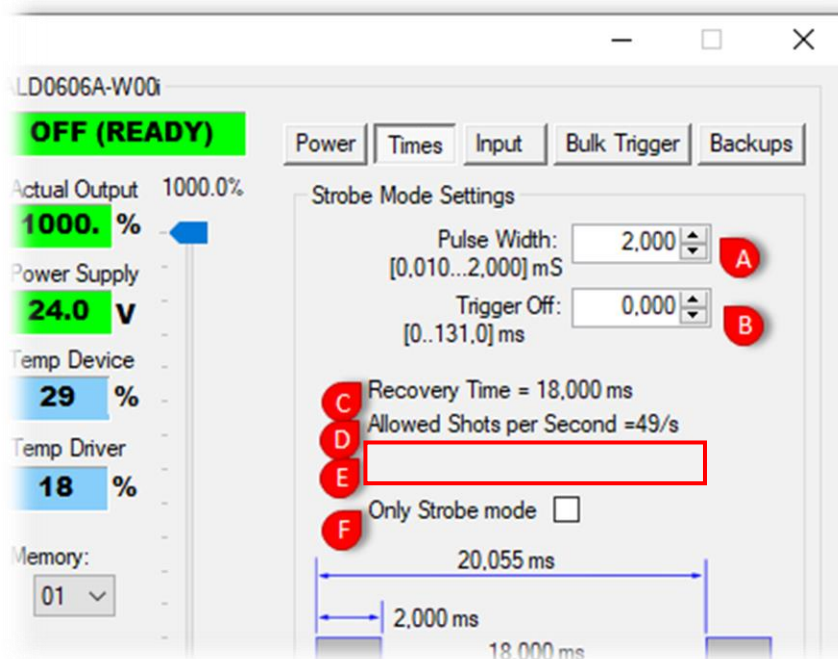
D. Allowed Shots per second

It is the maximum number of strobe operations that can be performed with the current times configuration.

E. Actual Trigger Off

Is the minimum time that the light will be off after a strobe operation. If Trigger Off is greater than the Recovery Time value then Actual Trigger Off will be set to the Trigger Off value stated.

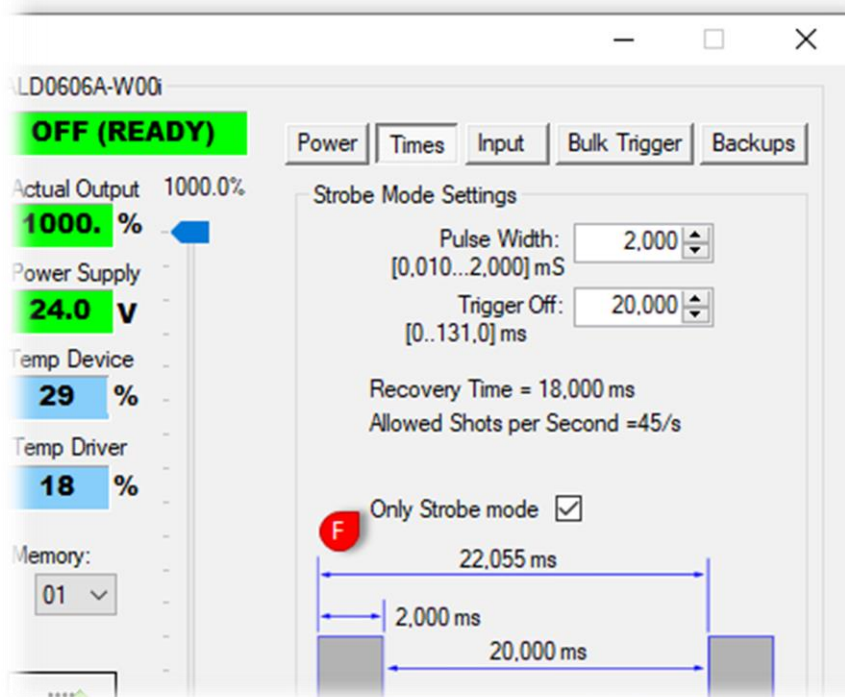
If this label is not shown, it means that the Actual Trigger Off is the Trigger Off value.



F. Only Strobe mode

When checked the lighting will only operate in strobe mode (FLASHING) regardless the width of the input trigger signal.

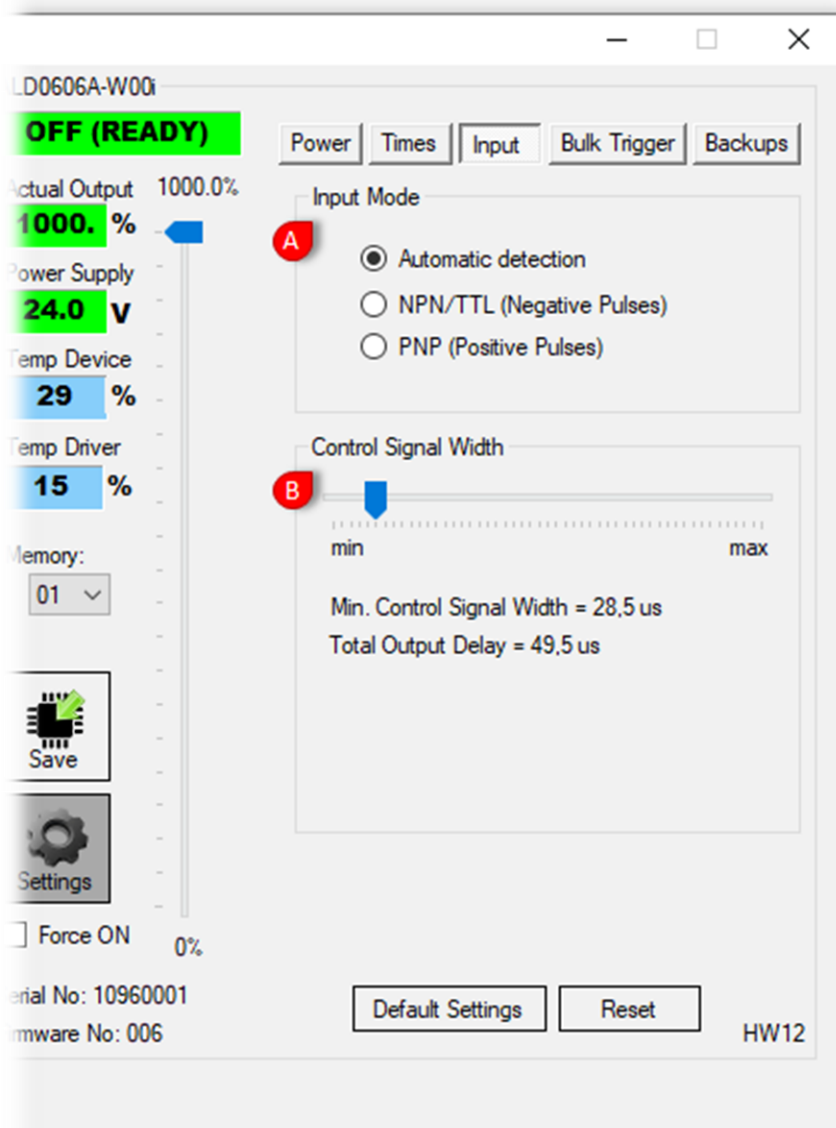
Max. Powered Time control will be disabled as the device will not work in this mode.



G. Max Powered Time

Sets maximum time that the device can operate in powered mode (POWERED).

5.2.3. Input Tab



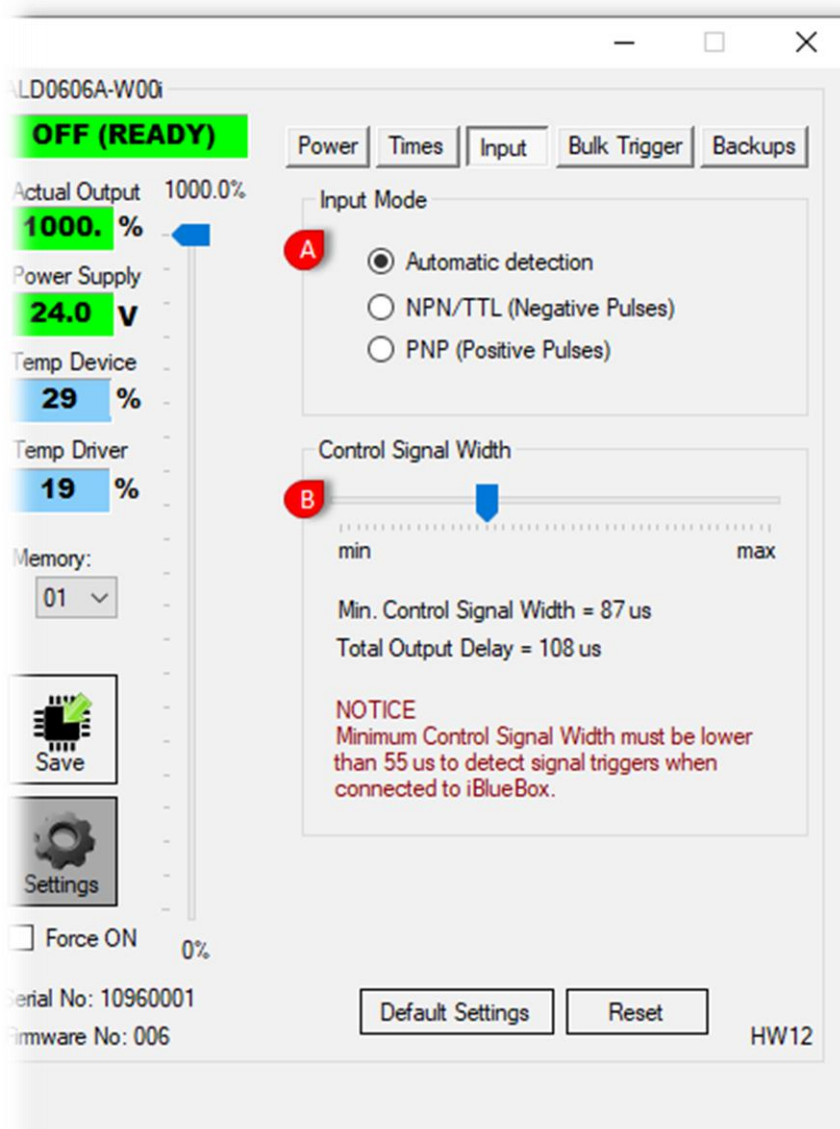
A. Input mode

This parameter allows to select which type of pulses will trigger the device. It can be set to:

- **Automatic detection:** the pulse type that will trigger the light will be the same that the first pulses type received. For example if the first received pulses are NPN then the device will only be affected by NPN pulses from then on.
- **NPN/TTL (Negative pulses):** the device will only take into account negative pulses.
- **PNP (Positive pulses):** the device will only take into account positive pulses.

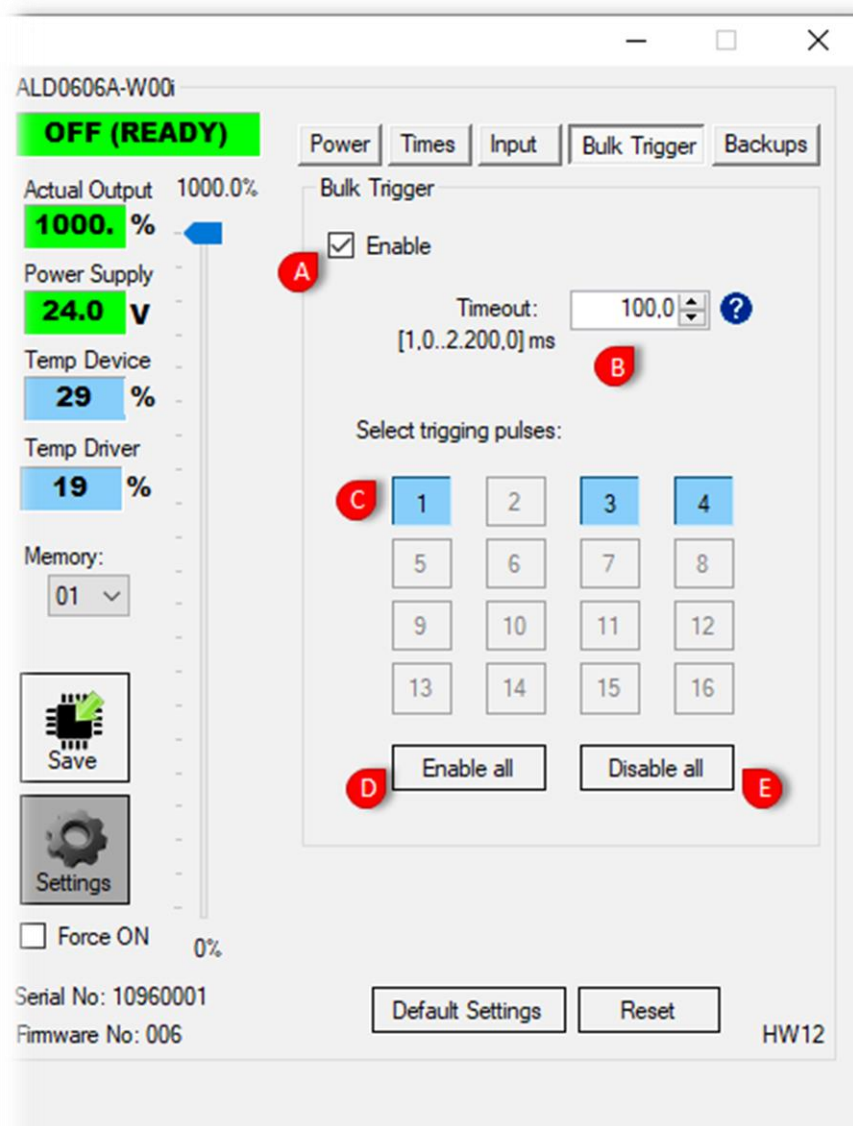
B. Control Signal Width

This parameter sets the minimum wide that the control signal pulses must be to be considered a valid pulse. This can be very useful in high electric noise environments, acting as an input filter.



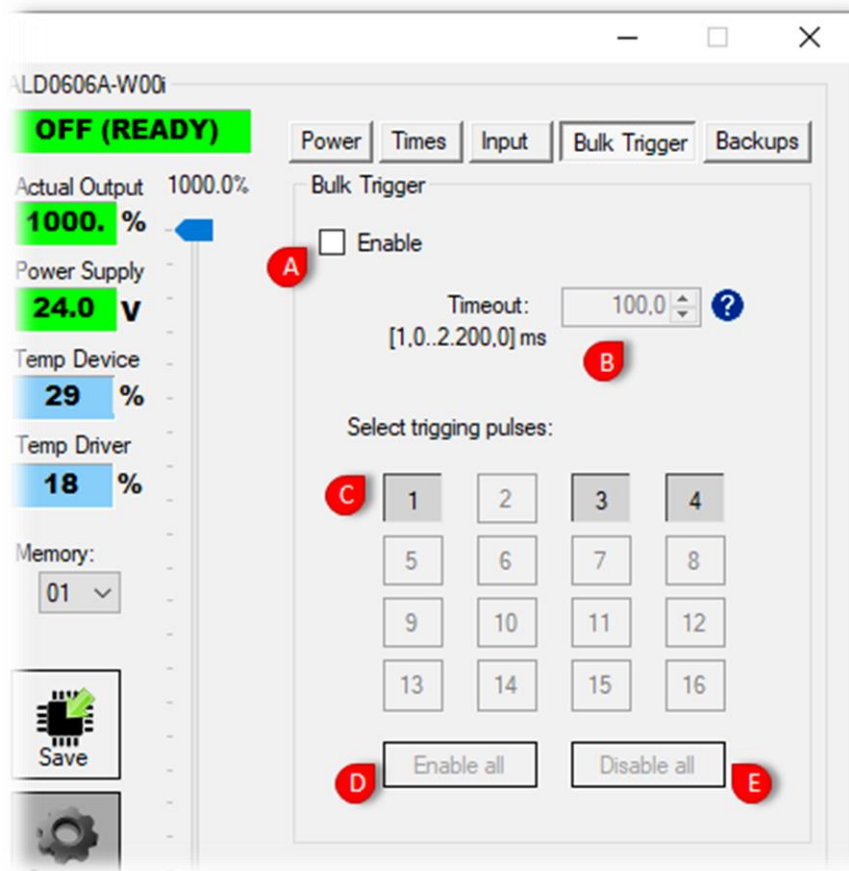
When the Minimum Control Signal Width is set to a value higher than 55us a warning message will be shown. Any value can be set but if you are using an iBlueBox to trigger a device its Minimum Control Signal Width must be lower than 55us. Otherwise the device will discard the trigger coming from the iBlueBox.

5.2.4. Bulk Trigger tab

*A. Enable*

When checked the Bulk Trigger parameters will be taken into account.

When unchecked Bulk Trigger parameters will be ignored and the device will operate normally.



B. *Timeout*

This parameter is the minimum time between two pulses for a device pulses counter reset.

C. *Triggering pulses*

Each iBlueDrive2 device can be configured to be turned on at up to 16 trigger points of a pulse train. This means that it is possible to select in which pulses the device will turn on and in which pulses it will remain off.

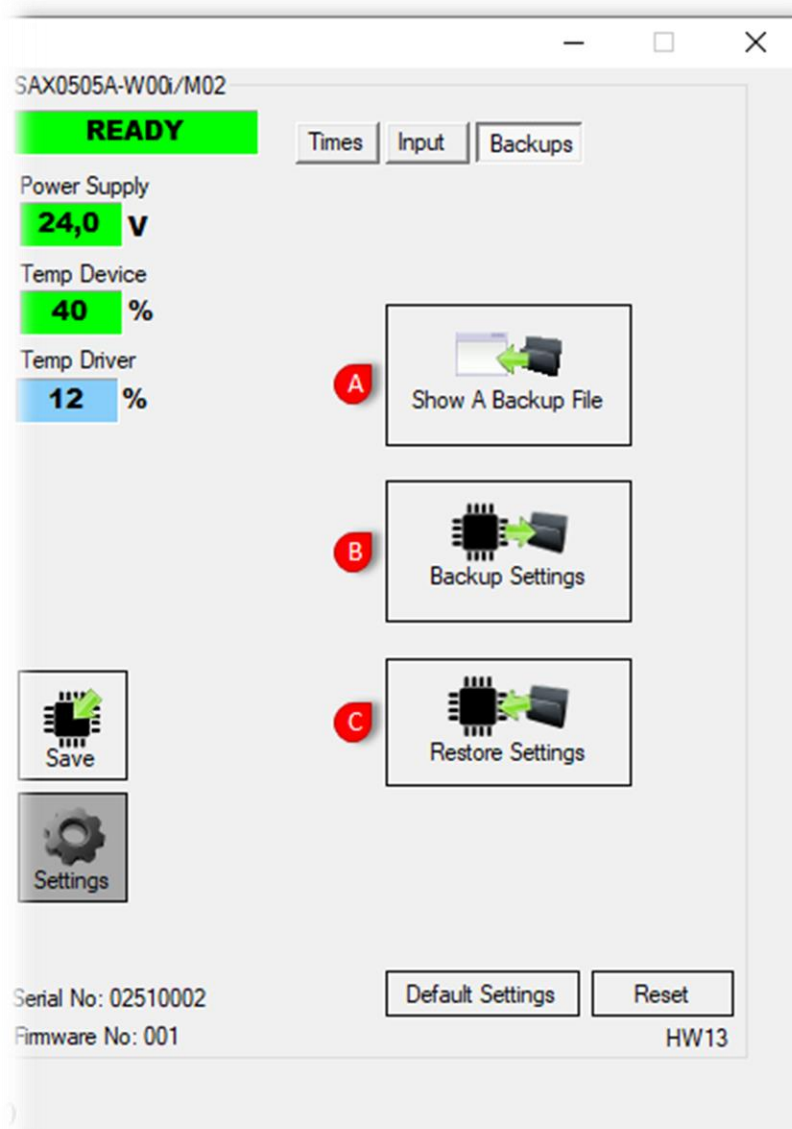
D. *'Enable all' button*

When clicked automatically enables all the triggering pulses.

E. *'Disable all' button*

When clicked automatically disables all the triggering pulses.

5.2.5. Backups tab



A. Show a backup file

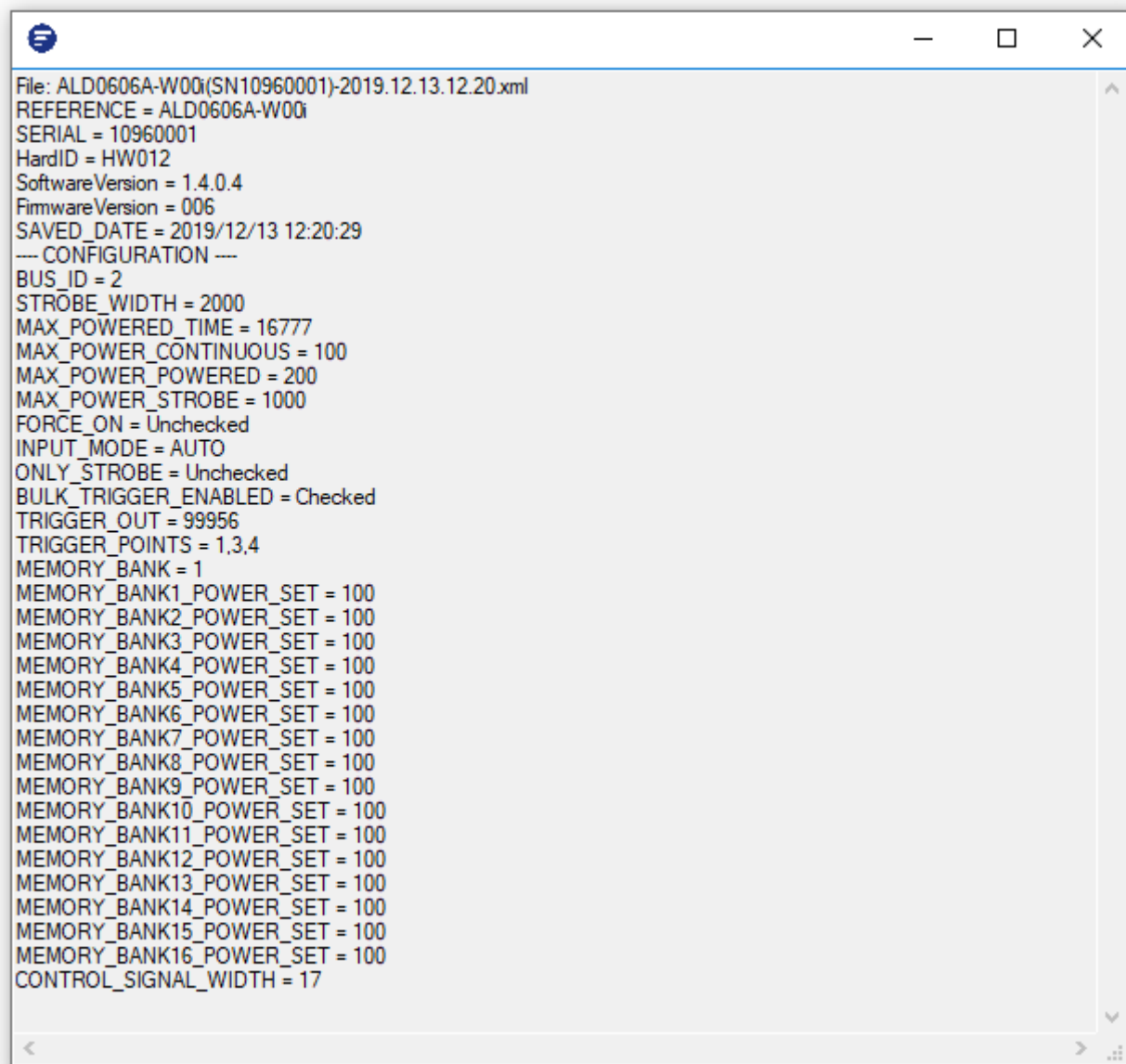
This button will pop a file explorer window which allows the user to select a previously generated backup file and check the parameters saved.

B. Backup settings

This feature allows to save the current configuration parameters in a file that can be checked and restored later. You will be asked for the folder and file name when this button is pressed.

C. Restore settings

This button will pop a file explorer window which allows the user to select a previously generated backup file and dump the parameter values saved to the currently selected device.

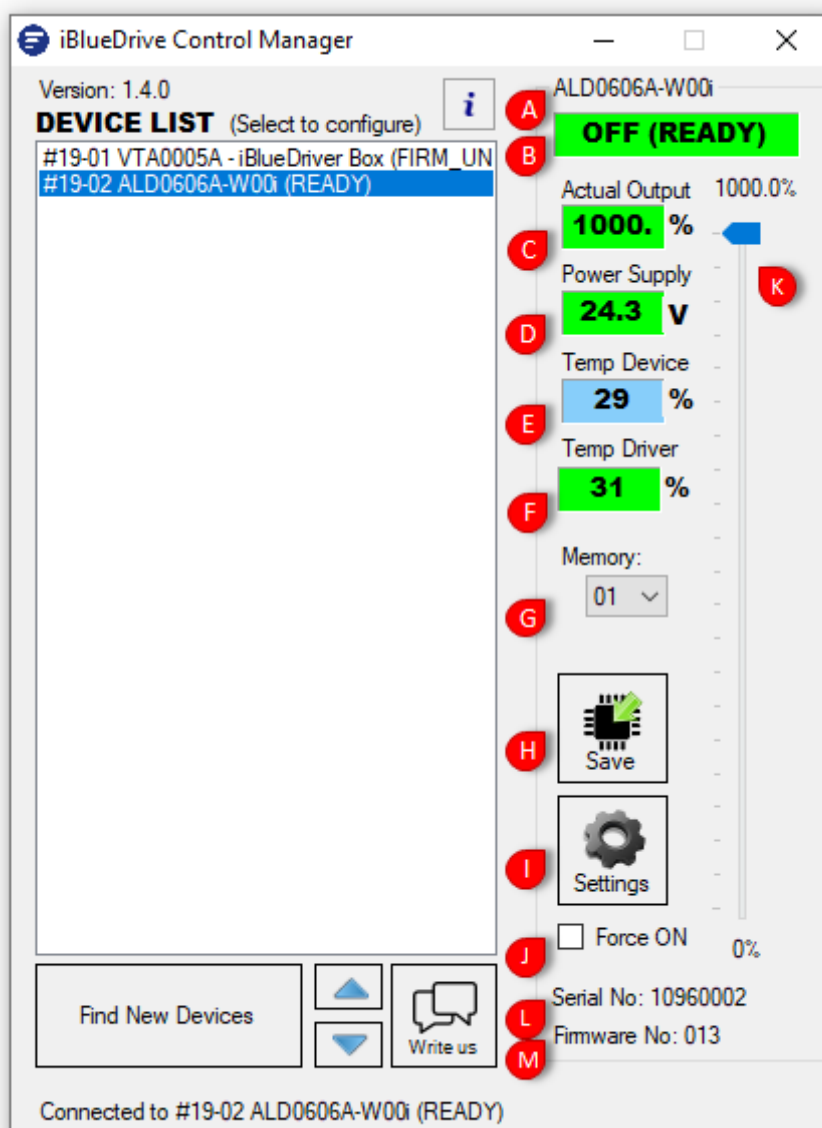


6. iBlueDrive lightings

The following sections show the different controls and screens of the 1.4.0 iBlueDrive Control Manager version when configuring an iBlueDrive device.

6.1. Main View

This view will be shown when clicking on the device designator.



A. Product name label

Shows the product name of the device

B. Lighting Status

- **OFF (READY):** the device light is off.
- **CONT.:** the device light is in continuous mode.
- **POWERED:** the device light is in powered mode.
- **FLASHING:** the device light is firing strobe pulses.
- **LOW VOLTAGE:** the input voltage is lower than 18.3V
- **OVERHEAT:** the driver temperature or the device temperature is higher than the operation temperature.

C. Actual Output

Lighting output power related to the maximum power in continuous mode, expressed as a percentage.

D. Power Supply

Voltage of the input power supply.

E. Device temperature

Temperature of the device related to its maximum operation temperature, expressed as a percentage.

F. Driver temperature

Temperature of the driver related to its maximum operation temperature, expressed as a percentage.

G. Memory

Indicates de memory bank currently in use. iBlueDrive lightings counts with 16 memory banks. Each memory bank can record one power output value.

H. *Save Button*

Saves the actual parameters configuration in the device's RAM memory. The button border turns red when some parameter's value has changed showing that the RAM memory is out of date. If the power supply is then disconnected before saving those changes will be lost.



I. *Settings Button*

Opens the side panel to operate the advanced settings

J. *Force ON*

When checked the lighting will turn on regardless the trigger input or the control signal, when unchecked the lighting will turn on according to the trigger input or the control signal.

K. *Power Slider*

Controls the lighting output power.

L. *Serial No*

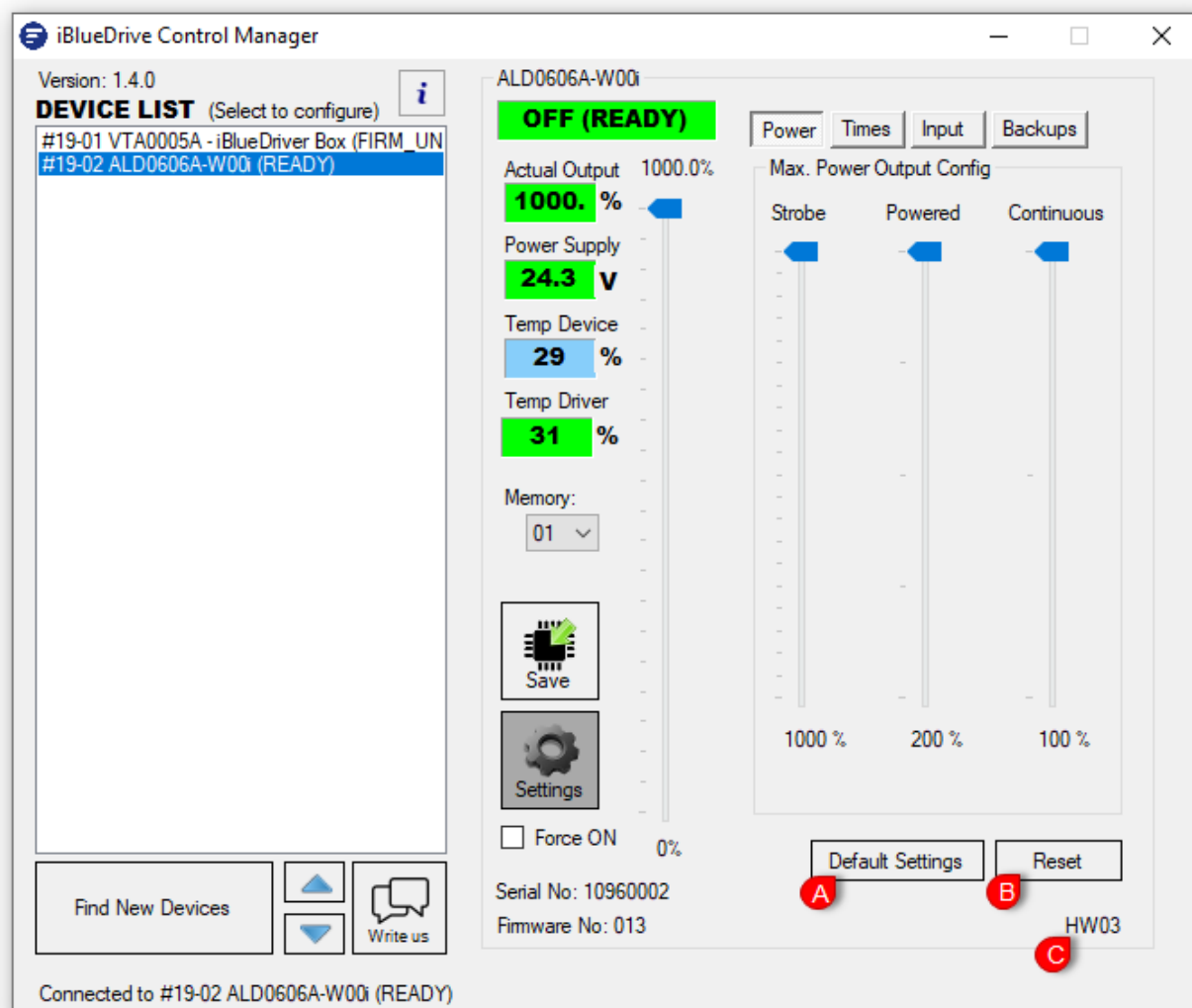
Shows the serial number. This number is unique for every single DCM device.

M. *Firmware*

Shows the firmware version of the device.

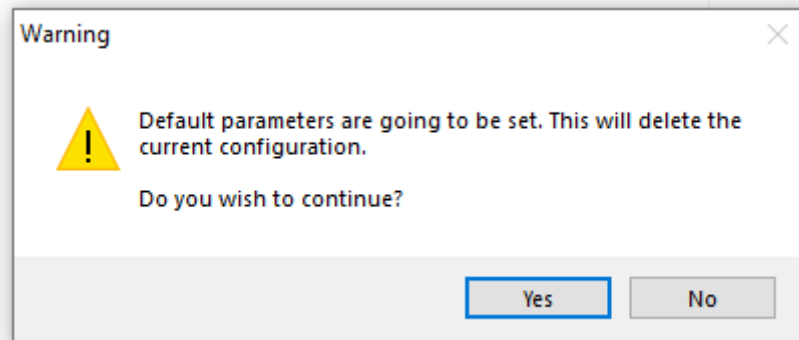
6.2. Settings

The configuration panel displays all configurable parameters of the device, allowing full adjustment to suit the needs of the application.



A. Default Settings button

Set all the parameters to their default value and save them in the device's RAM memory. A confirmation message will be shown before proceed.



B. Reset Button

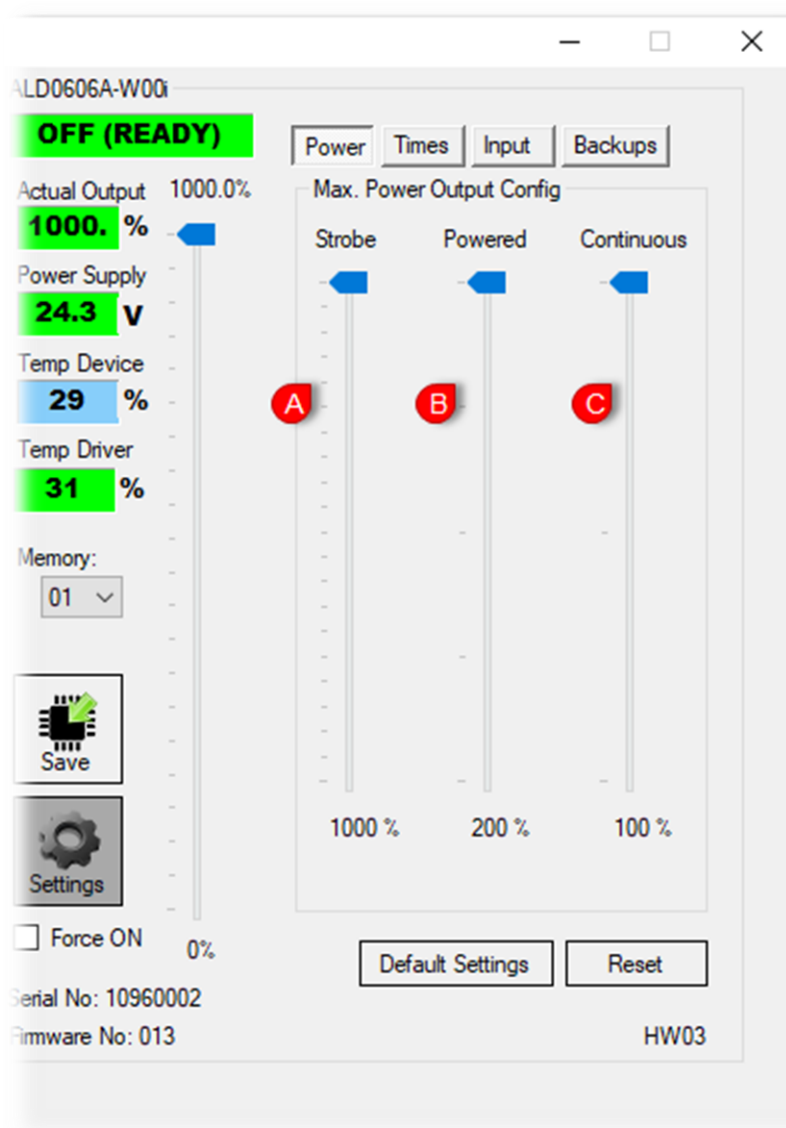
Performs a reset in the device. This equals to disconnect and then reconnect the device to the power supply.

C. Hardware label

Shows the hardware version of the device electronics. For the iBlueDrive is HW003.

6.2.1. Power tab

This tab shows the controls related to the output power of the device.



A. Strobe slider

Controls the device output power when working in strobe mode (FLASHING).

B. Powered slider

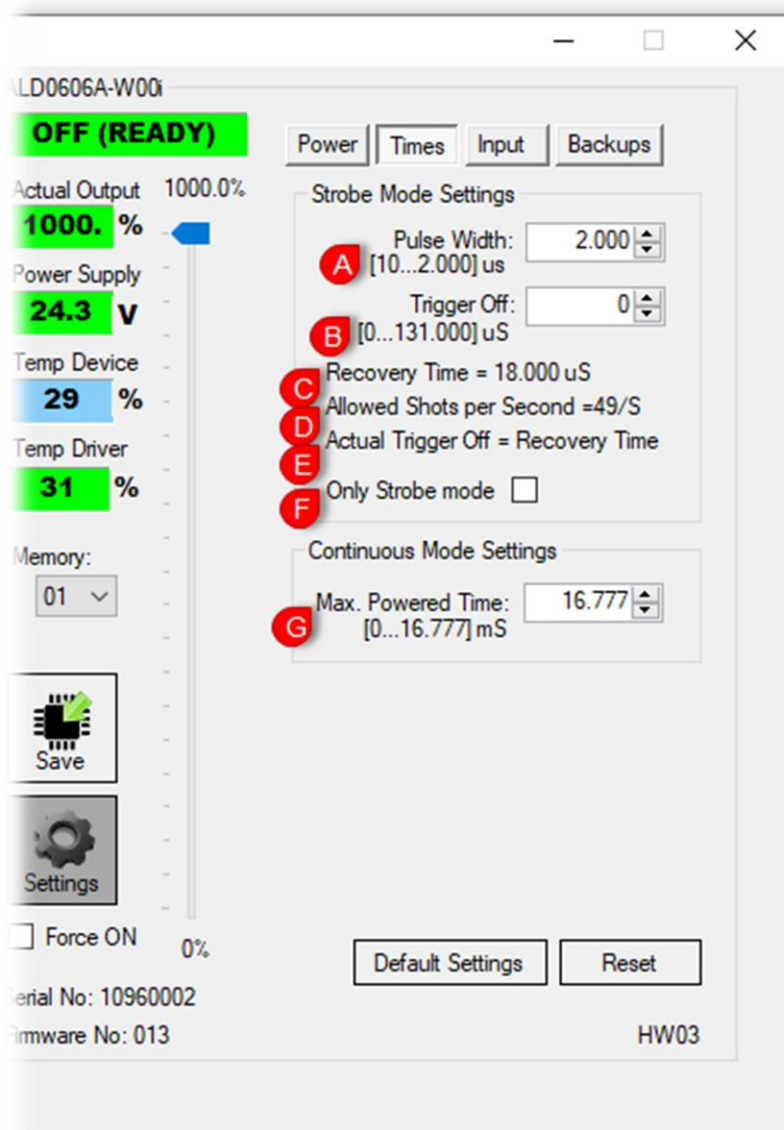
Controls the device output power when working in powered mode (POWERED).

C. Continuous slider

Controls the device output power when working in continuous mode (CONT.).

6.2.2. Times Tab

This tab shows the controls related to the signal timing configuration.



A. Pulse Width

Allows you to change the light on time when operating in strobe mode (FLASHING).

B. Trigger Off

Allows you to customize the light off time after a strobe operation.

C. Recovery Time

Is the minimum time that the light must be off after a strobe operation. The recovery time will depend on the Pulse Width value.

The relation between Recovery Time and Pulse Width cannot be modified as it depends on technical and constructive parameters specific to each product.

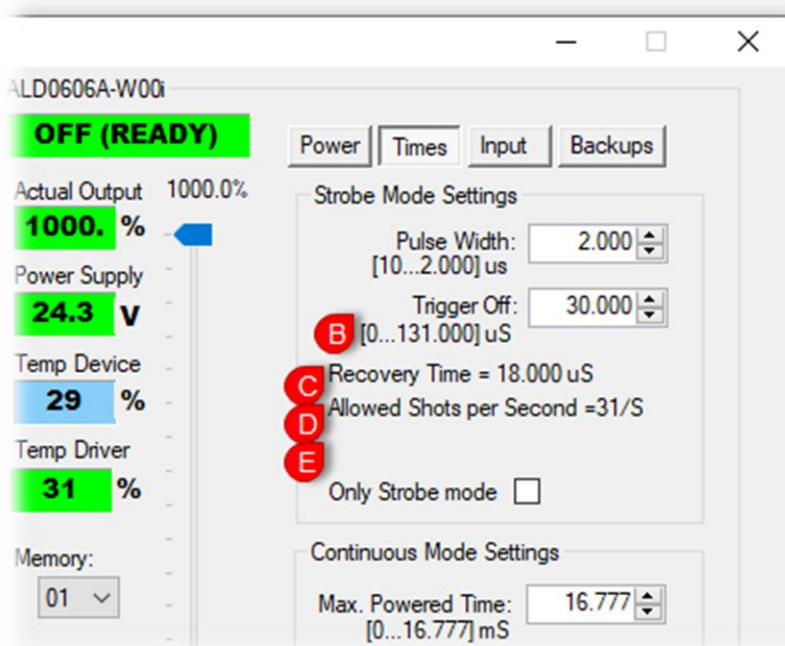
D. Allowed Shots per second

It is the maximum number of strobe operations that can be performed with the current times configuration.

E. Actual Trigger Off

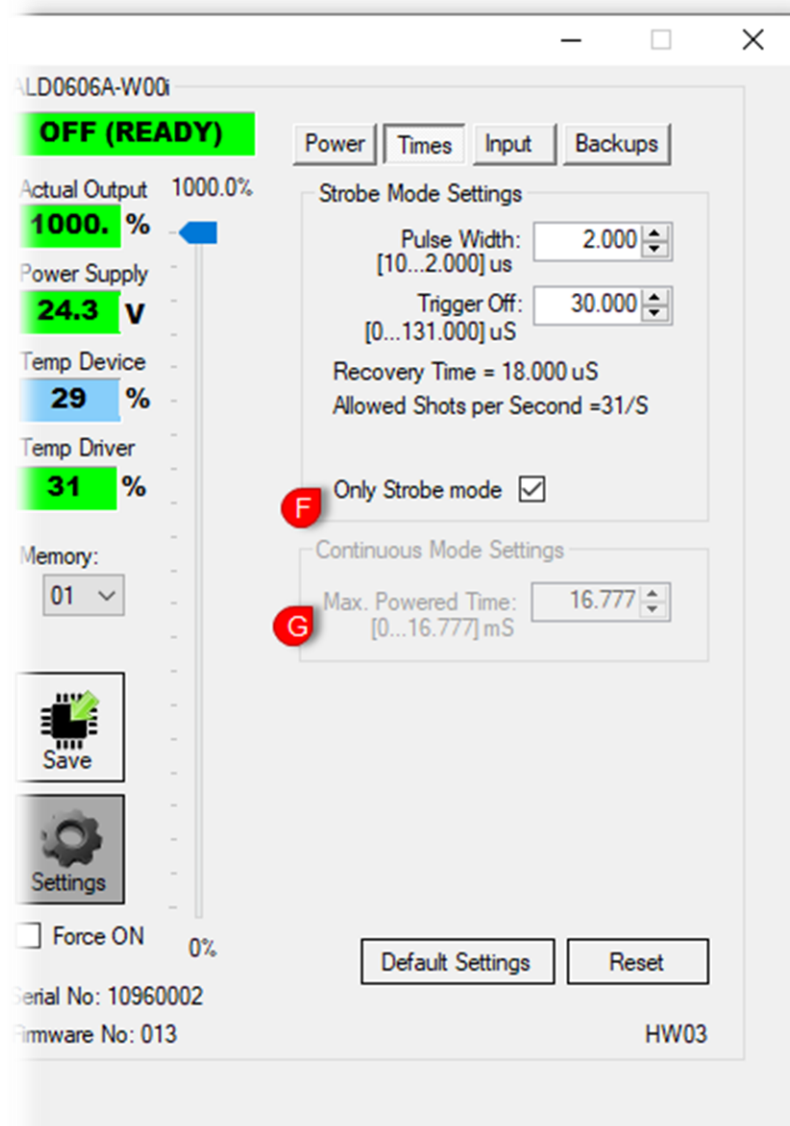
Is the minimum time that the light will be off after a strobe operation. If Trigger Off is greater than the Recovery Time value then Actual Trigger Off will be set to the Trigger Off value stated.

If this label is not shown means that the Actual Trigger Off is the Trigger Off value.



F. Only Strobe mode

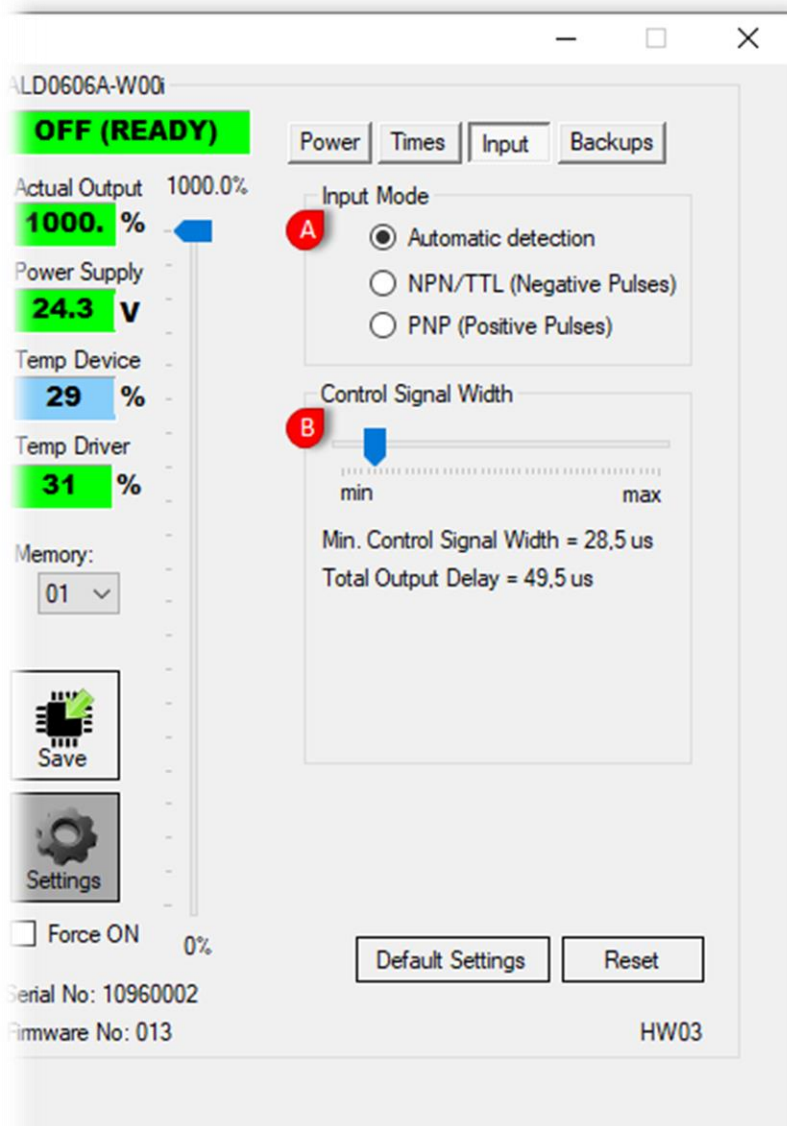
When checked the lighting will only operate in strobe mode (FLASHING) regardless the width of the input trigger signal.



G. Max Powered Time

Sets maximum time that the device can operate in powered mode (POWERED).

6.2.3. Input Tab



A. Input mode

This parameter allows to select which type of pulses will trigger the device. It can be set to:

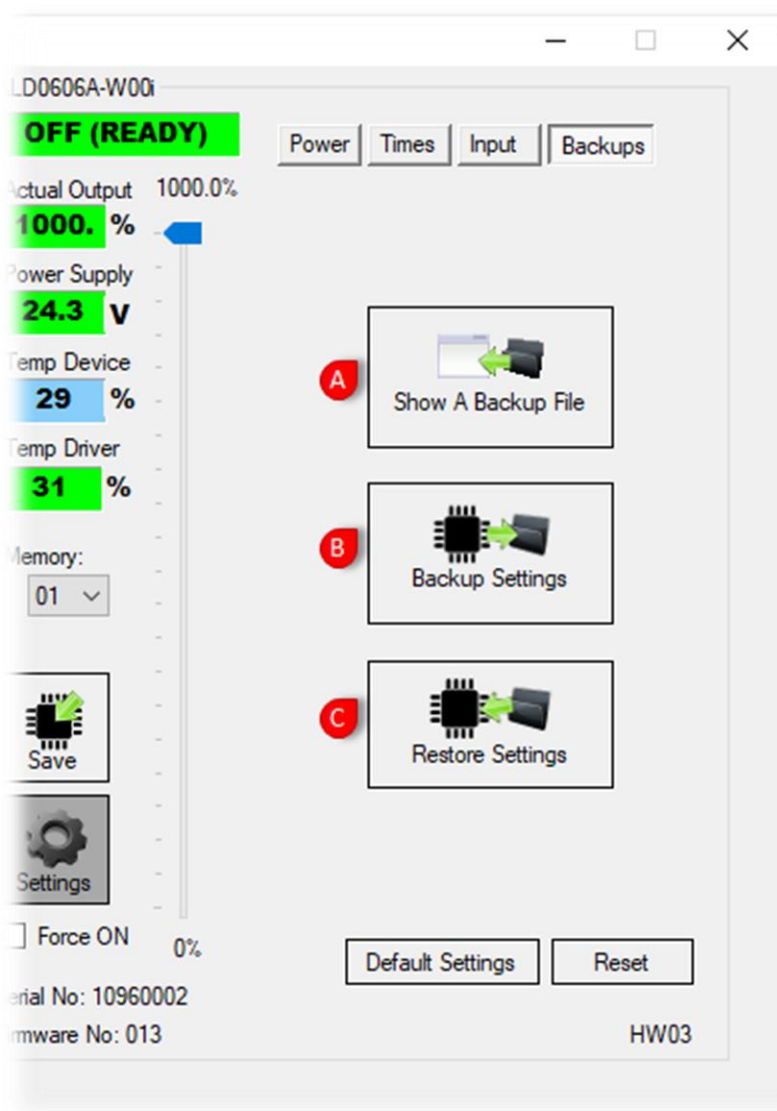
- **Automatic detection:** the pulse type that will trigger the light will be the same that the first pulses type received. For example, if the first received pulses are NPN then the device will only be affected by NPN pulses from then on.
- **NPN/TTL** (Negative pulses): the device will only take into account negative pulses.
- **PNP** (Positive pulses): the device will only take into account positive pulses.

B. Control Signal Width

This parameter sets the minimum wide that the control signal pulses must be to be considered a valid pulse. This can be very useful in high electric noise environments, acting as an input filter. The value (in microseconds) is shown below the horizontal control bar.

The parameter “Total Output Delay” is the time between the trigger and the device turning on.

6.2.4. Backups tab



A. *Show a backup file*

This button will pop a file explorer window which allows the user to select a previously generated backup file and check the parameters saved.

B. *Backup settings*

This feature allows to save the current configuration parameters in a file that can be checked and restored later. You will be asked for the folder and file name when this button is pressed.

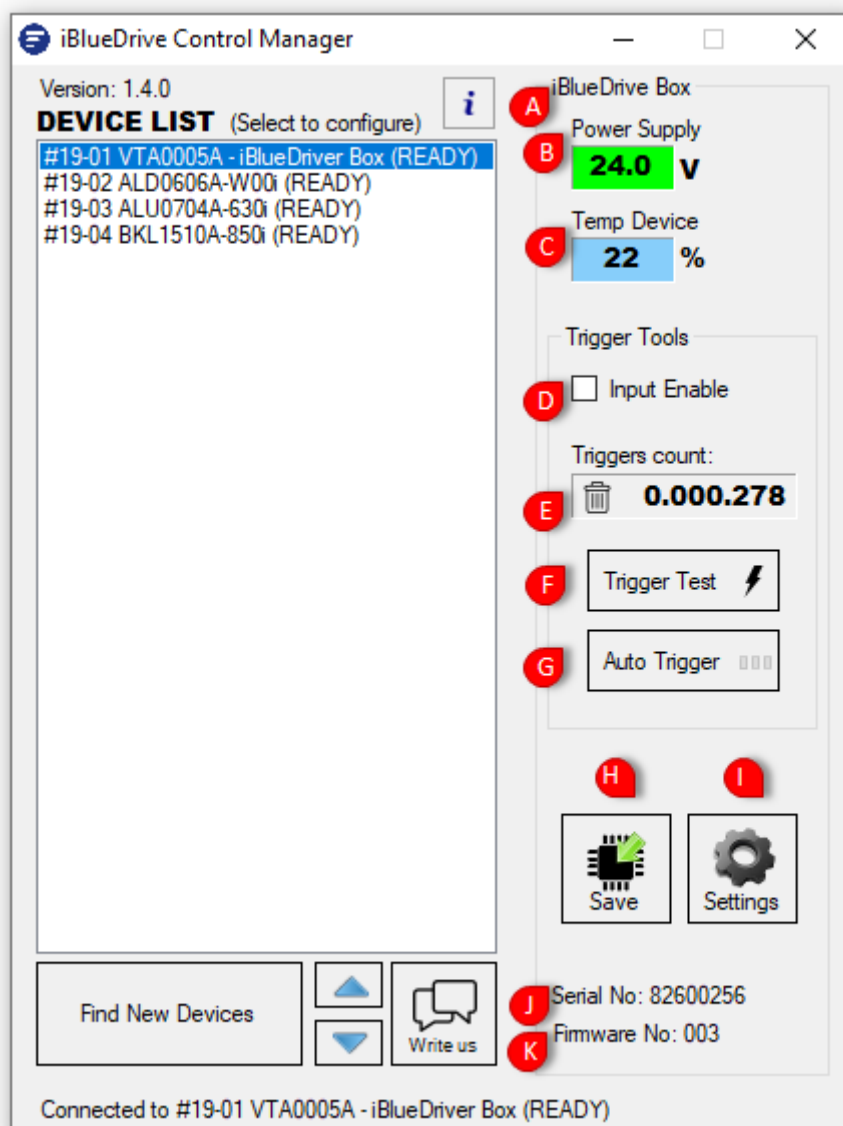
C. *Restore settings*

This button will pop a file explorer window which allows the user to select a previously generated backup file and dump the parameter values saved to the currently selected device.

7.

7. iBlueBox

7.1. Main View



A. *Product name label*

Shows the product name of the device

B. *Power Supply*

Voltage of the input power supply.

C. *Device temperature*

Temperature of the device related to its maximum operation temperature, expressed as a percentage.

D. *Input Enable*

When checked will ignore trigger input pulses.

E. *Trigger count*

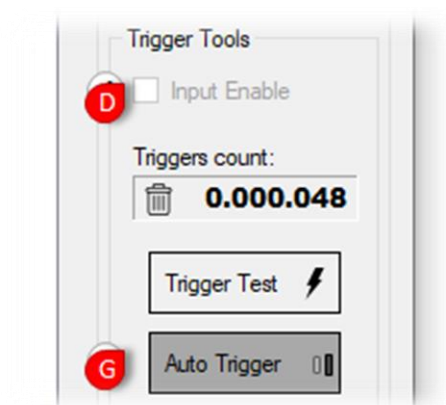
A counter that shows the number of pulses or pulse trains generated. It can be reset clicking on the trash icon.

F. *Trigger Test*

When clicked will make the iBlueBox generate a single pulse or pulse train (as configured in 'Output Setup' tab)

G. *AutoTrigger*

When activated will make the iBlueBox generate a pulse or pulse train (as configured in 'Output Setup' tab) repeatedly. Input enable will be ignored and, therefore, disabled.



H. *Save Button*

Saves the actual parameters configuration in the device's RAM memory. When the button border turns red means that some parameter's value has changed and the RAM memory is out of date. If the power supply is then disconnected those changes will be lost.

I. *Settings Button*

Opens the side panel to operate the advanced settings.

J. *Serial No*

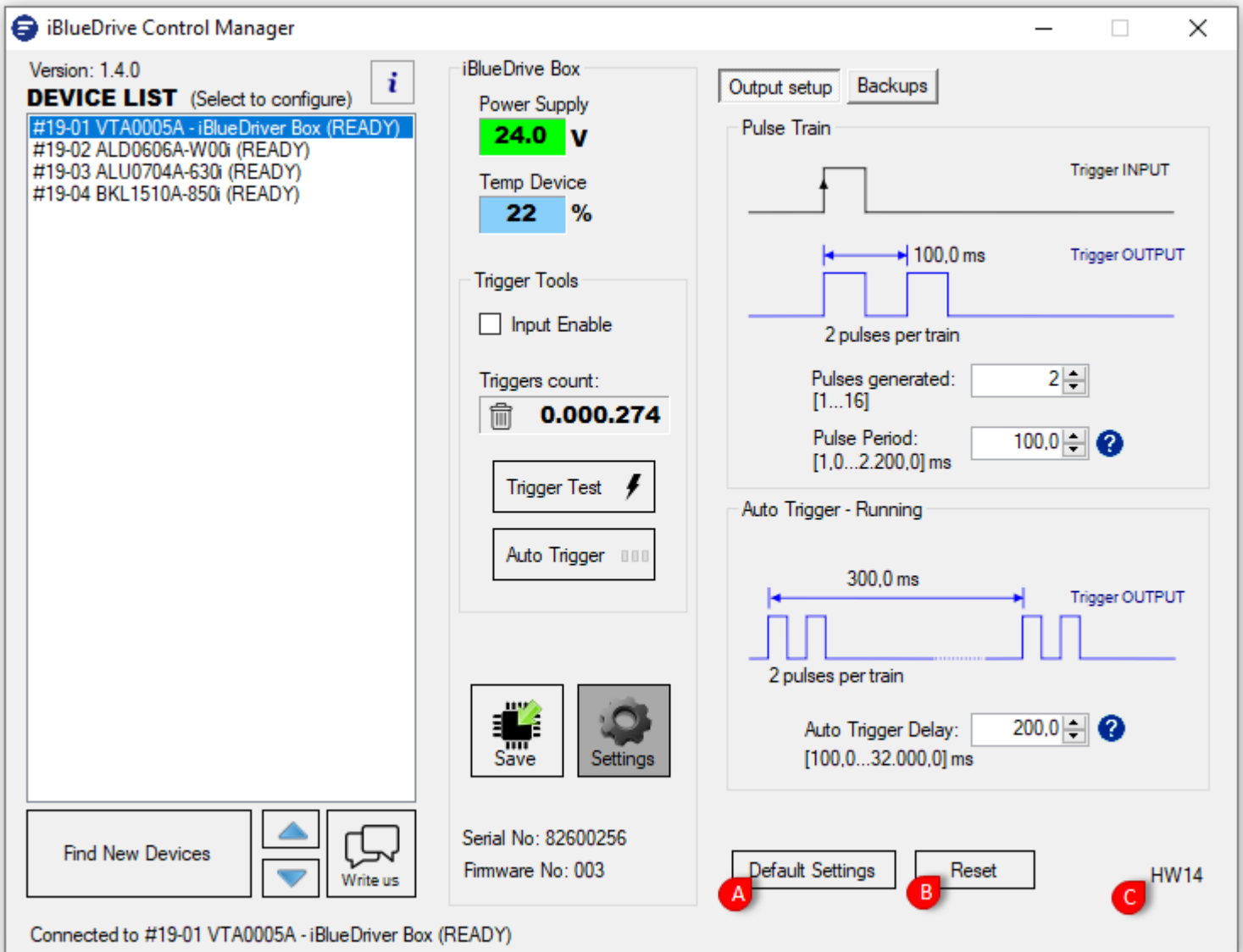
Shows the serial number. This number is unique for every single DCM device.

K. *Firmware*

Shows the firmware version of the device.

7.2. Settings

The configuration panel displays all configurable parameters of the device, allowing full adjustment to suit the needs of the application.



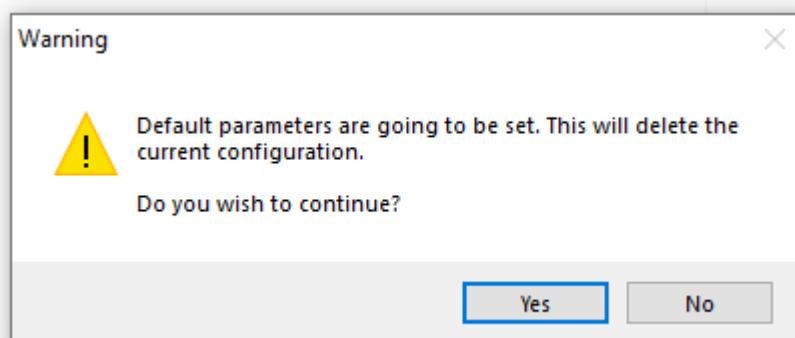
The screenshot displays the iBlueDrive Control Manager software interface. The window title is "iBlueDrive Control Manager". The version is 1.4.0. The interface is divided into several sections:

- DEVICE LIST (Select to configure):** A list of four devices:
 - #19-01 VTA0005A - iBlueDriver Box (READY)
 - #19-02 ALD0606A-W00i (READY)
 - #19-03 ALU0704A-630i (READY)
 - #19-04 BKL1510A-850i (READY)
- iBlueDrive Box:**
 - Power Supply: 24.0 V
 - Temp Device: 22 %
- Trigger Tools:**
 - ☐ Input Enable
 - Triggers count: 0.000.274
 - Trigger Test (lightning bolt icon)
 - Auto Trigger (four small squares icon)
- Buttons:**
 - Find New Devices
 - Write us (speech bubble icon)
 - Save (chip icon)
 - Settings (gear icon)
- Serial No:** 82600256
- Firmware No:** 003
- Output setup / Backups:**
 - Pulse Train:**
 - Trigger INPUT: A single high pulse.
 - Trigger OUTPUT: A pulse train with a period of 100.0 ms and 2 pulses per train.
 - Pulses generated: 2 (range [1...16])
 - Pulse Period: 100.0 ms (range [1,0...2.200,0] ms)
 - Auto Trigger - Running:**
 - Trigger OUTPUT: A pulse train with a period of 300.0 ms and 2 pulses per train.
 - Auto Trigger Delay: 200.0 ms (range [100,0...32.000,0] ms)
- Buttons at the bottom:**
 - Default Settings (labeled A)
 - Reset (labeled B)
 - HW14 (labeled C)

At the bottom left, it says "Connected to #19-01 VTA0005A - iBlueDriver Box (READY)".

A. Default Settings button

Set all the parameters to their default value and save them in the device's RAM memory. A confirmation message will be shown before proceed.



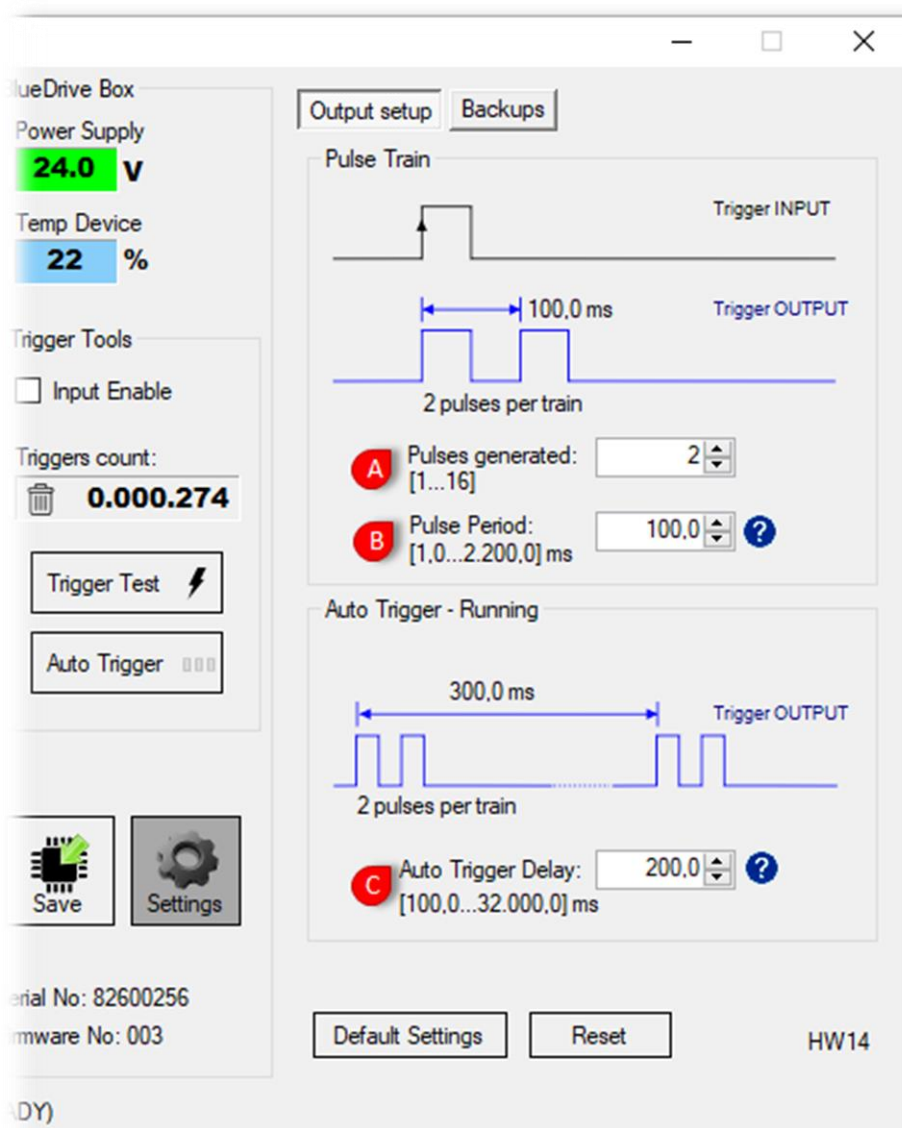
B. Reset Button

Performs a reset in the device. This equals to disconnect and then reconnect the device to the power supply.

C. Hardware label

Shows the hardware version of the device electronics. For the iBlueBox is either HW01 or HW14.

7.2.1. Output Setup Tab



D. Pulses generated

This feature only has effect when the Input enable checkbox is checked.

Sets the number of pulses or pulse trains that the iBlueBox will generate through the TRIGGER OUPUT every time that a pulse is received in the TRIGGER INPUT or every period of Auto Trigger when it is on.

E. Pulse period

Sets the time between two consecutive pulses of a pulse train

F. Auto Trigger Delay

This control only has effect when the Auto Trigger feature is enabled.

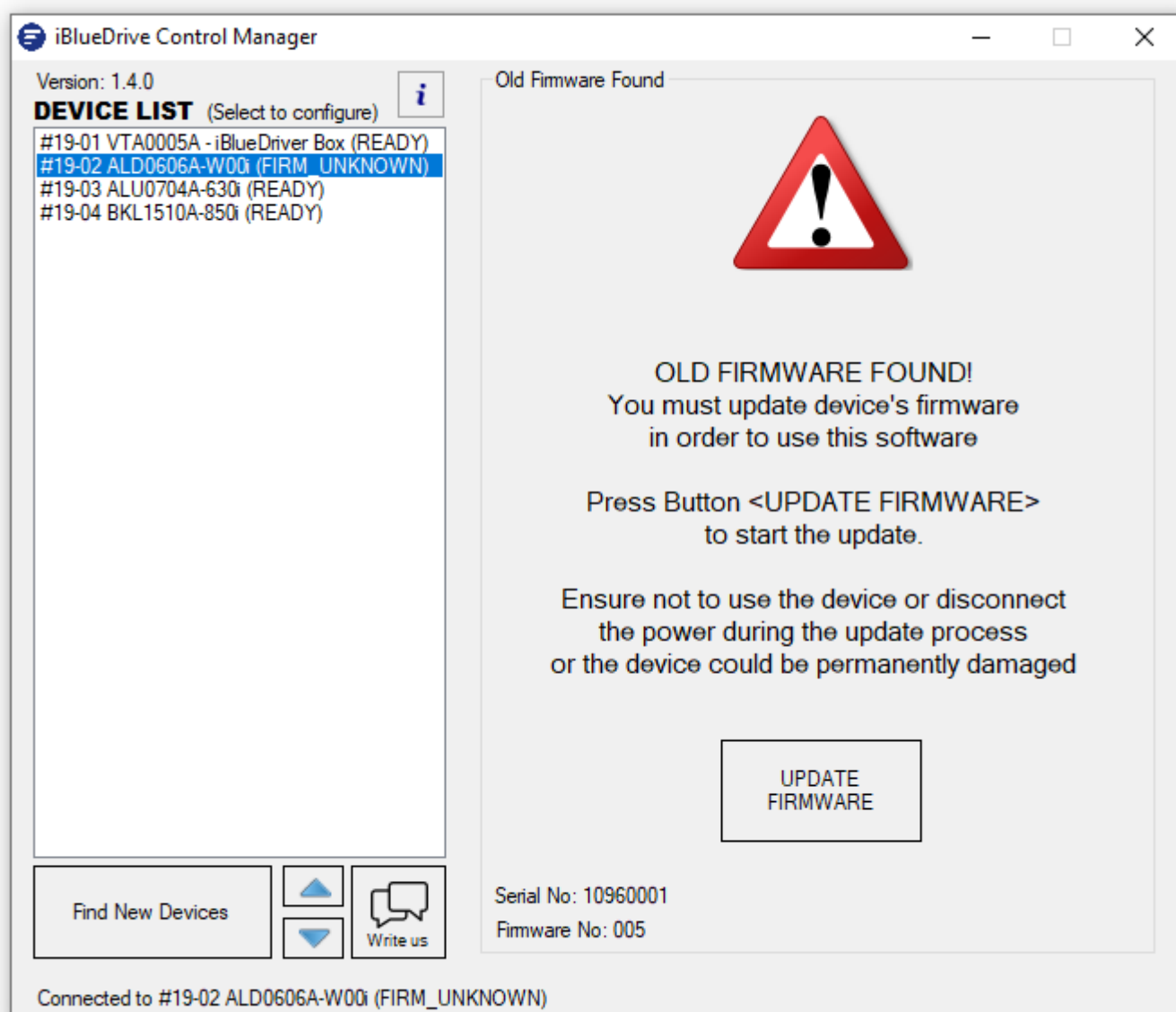
Sets the time between two train pulses.

8. Firmware update

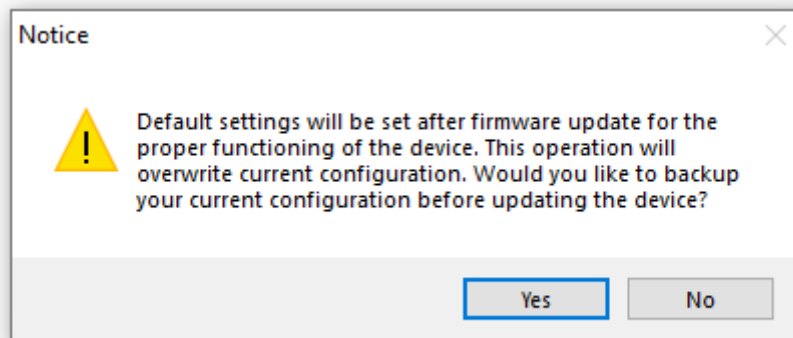
iBlueDrive technology is continually being improved. New features are periodically added to both the device firmware and the iBlueDrive Control Manager software. It is important that the devices are updated in order to benefit from their full potential. Each firmware version works with certain software versions.

8.1. Firmware update

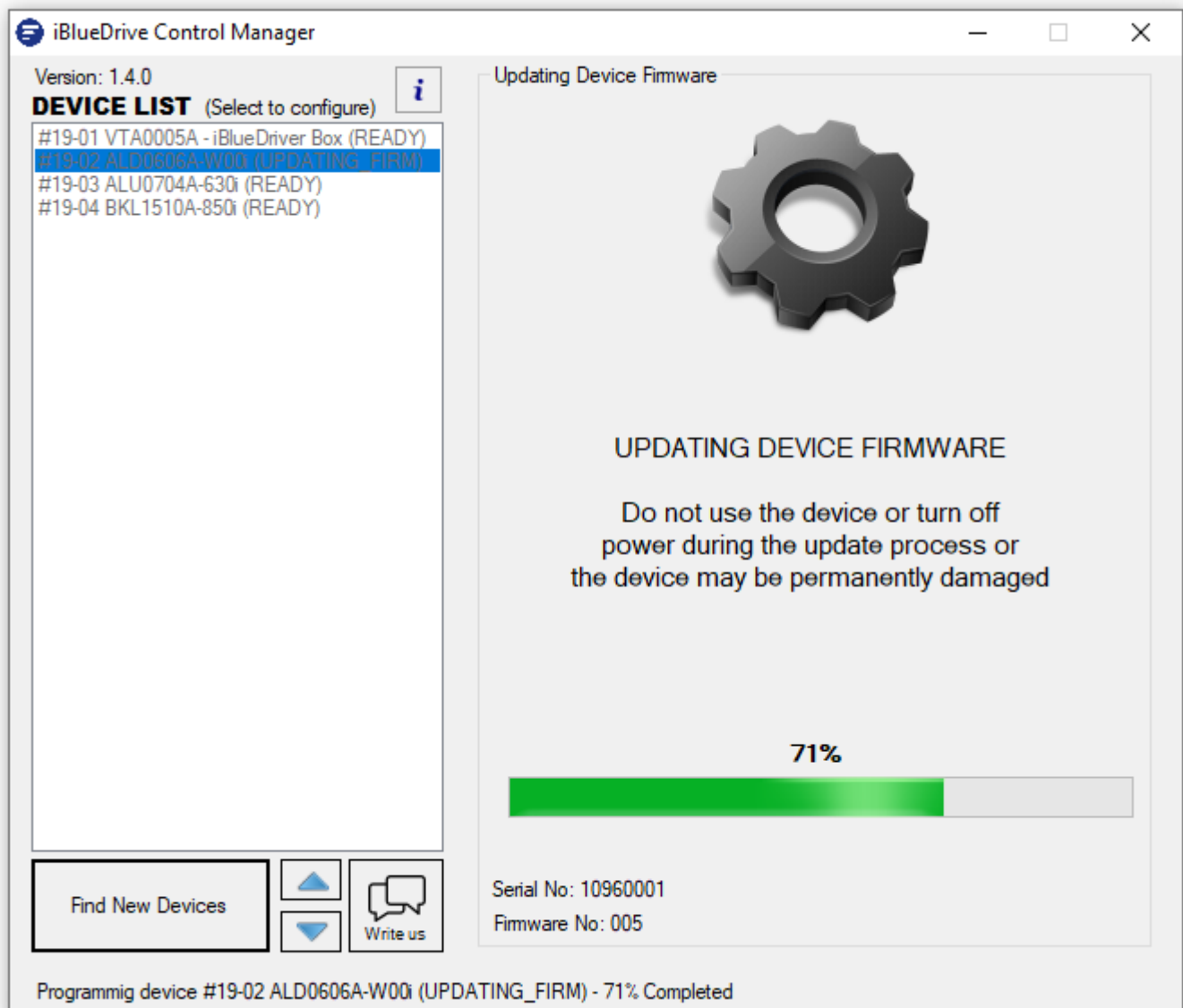
When a more modern software version than the firmware version is used, the device must be updated in order to be controlled from a PC. The following message will appear when this situation takes place. Notice that the device's status is FIRM_UNKNOWN.



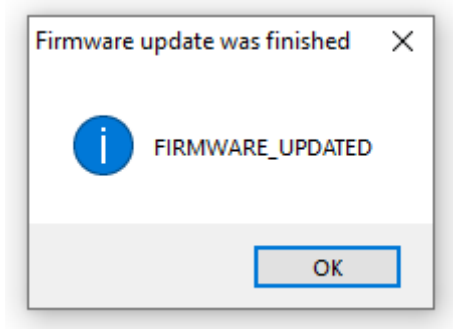
When the 'UPDATE FIRMWARE' is clicked, a popup message will ask for backup the current device configuration. You can skip this step clicking 'No' or backup your data clicking on 'Yes'.



The updating process then will start.



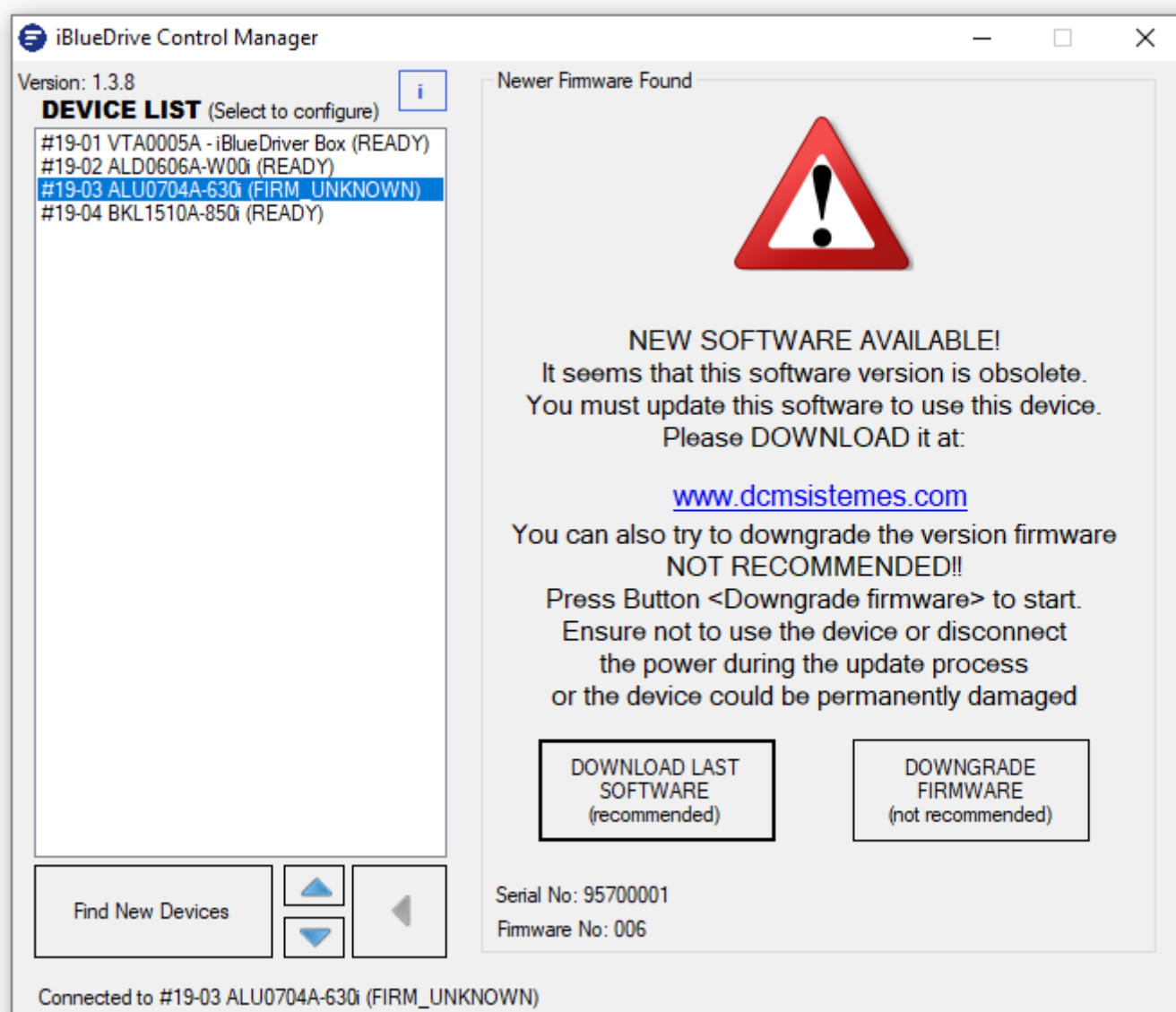
Finally, this message will appear showing that the updating process has been successful.

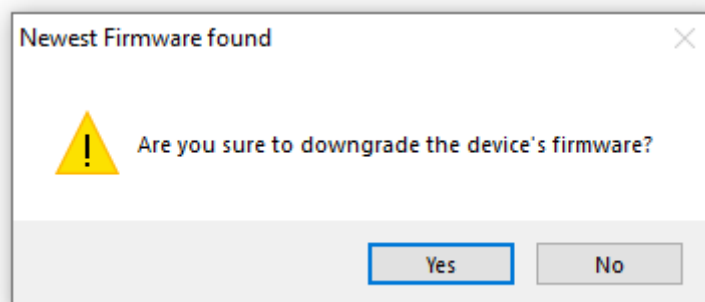


8.2. Firmware downgrade

When a less modern software version than the firmware version is used the user can choose between downgrading the connected device (not recommended) or updating the iBlueDrive Control Manager Software.

The following message will appear when this situation takes place. Notice that the device's status is FIRM_UNKNOWN.





8.3. Device Reference Unknown

The 'devices.dat' file is needed for operate the devices. This file is included in the installation package. As new products are continually being developed also this file need to be updated.

When a new product is not recognised by the software its name and state will become "UNDEF (REF_UNKNOWN)" and you will be asked to update the 'devices.dat' file. This can be done in two ways: by clicking the "Download 'devices.dat' form internet" button, which will download the newest version of the file from the DCM web site (internet connection required), or by manually uploading the updated file.

The file can be downloaded from the DCM website

(<https://www.dcmsistemas.com/en/products/accessories/software/>)

